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(54) **3D UPPER GARMENT TRACKING**

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(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,604,843	A	2/1997	Shaw et al.
5,689,559	A	11/1997	Park

(Continued)

FOREIGN PATENT DOCUMENTS

CN	201986710	U	9/2011
CN	103049761	B	8/2016

(Continued)

OTHER PUBLICATIONS

"U.S. Appl. No. 17/301,690, Non Final Office Action mailed Jan. 19, 2024", 16 pgs.

(Continued)

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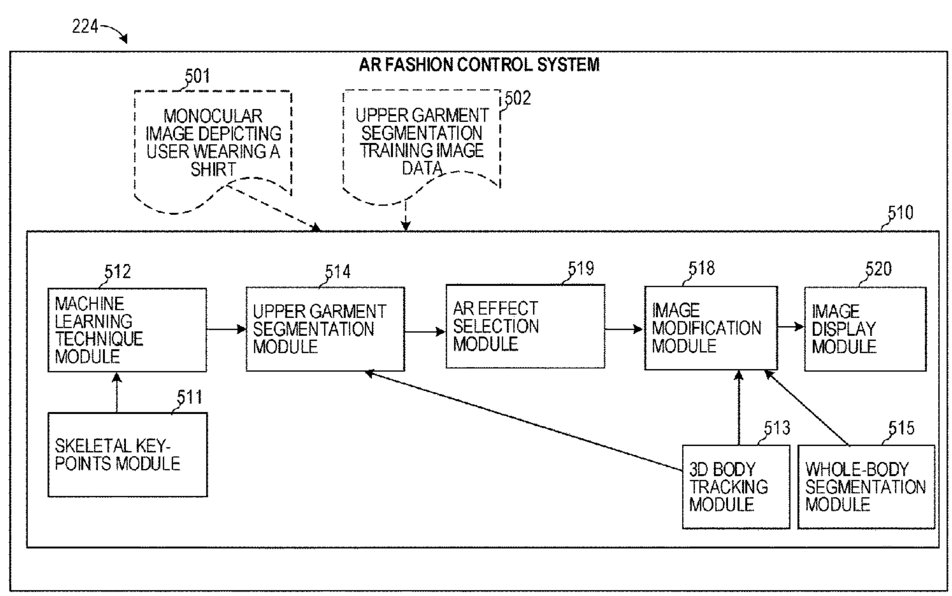
(57) **ABSTRACT**

Methods and systems are disclosed for performing operations comprising: receiving a video that includes a depiction of a person wearing a fashion item; generating a three-dimensional (3D) model of the person depicted in the video; generating a segmentation of the fashion item worn by the person depicted in the video; outlining a portion of the 3D model based on the segmentation of the fashion item; tracking movement of the portion of the 3D model in the video; and in response to tracking the movement of the portion of the 3D model in the video, modifying a display position of one or more augmented reality elements on the fashion item in the video.

20 Claims, 11 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

5,880,731	A	3/1999	Liles et al.
6,023,270	A	2/2000	Brush, II et al.
RE36,919	E	10/2000	Park
RE37,052	E	2/2001	Park
6,223,165	B1	4/2001	Lauffer
6,650,793	B1	11/2003	Lund et al.
6,772,195	B1	8/2004	Hatlelid et al.
6,804,417	B1	10/2004	Lund et al.
6,842,779	B1	1/2005	Nishizawa
7,308,332	B2	12/2007	Okada et al.
7,342,587	B2	3/2008	Danzig et al.
7,468,729	B1	12/2008	Levinson
7,636,755	B2	12/2009	Blattner et al.
7,639,251	B2	12/2009	Gu et al.
7,775,885	B2	8/2010	Van et al.
7,809,192	B2	10/2010	Gokturk et al.
7,859,551	B2	12/2010	Bulman et al.
7,885,931	B2	2/2011	Seo et al.
7,925,703	B2	4/2011	Dinan et al.
7,971,156	B2	6/2011	Albertson et al.
7,996,793	B2	8/2011	Latta et al.
8,088,044	B2	1/2012	Tchao et al.
8,095,878	B2	1/2012	Bates et al.
8,099,462	B2	1/2012	Sheng et al.
8,108,774	B2	1/2012	Finn et al.
8,117,281	B2	2/2012	Robinson et al.
8,130,219	B2	3/2012	Fleury et al.
8,146,005	B2	3/2012	Jones et al.
8,151,191	B2	4/2012	Nicol
8,275,590	B2	9/2012	Szymczyk et al.
8,359,247	B2	1/2013	Vock
RE43,993	E	2/2013	Park
8,379,920	B2	2/2013	Yang et al.
8,384,719	B2	2/2013	Reville et al.
RE44,054	E	3/2013	Kim
RE44,068	E	3/2013	Park
RE44,106	E	3/2013	Park
8,396,708	B2	3/2013	Park et al.
RE44,121	E	4/2013	Park
8,425,322	B2	4/2013	Gillo et al.
8,458,601	B2	6/2013	Castelli et al.
8,462,198	B2	6/2013	Lin et al.
8,484,158	B2	7/2013	Deluca et al.
8,487,938	B2	7/2013	Latta et al.
8,495,503	B2	7/2013	Brown et al.
8,495,505	B2	7/2013	Smith et al.
8,504,926	B2	8/2013	Wolf
8,559,980	B2	10/2013	Pujol
8,564,621	B2	10/2013	Branson et al.
8,564,710	B2	10/2013	Nonaka et al.
8,581,911	B2	11/2013	Becker et al.
8,582,918	B2	11/2013	Nakada et al.
8,597,121	B2	12/2013	del Valle
8,601,051	B2	12/2013	Wang
8,601,379	B2	12/2013	Marks et al.
8,606,645	B1	12/2013	Applefeld
8,632,408	B2	1/2014	Gillo et al.
8,648,865	B2	2/2014	Dawson et al.
8,659,548	B2	2/2014	Hildreth
8,683,354	B2	3/2014	Khandelwal et al.
8,692,830	B2	4/2014	Nelson et al.
8,810,513	B2	8/2014	Ptucha et al.
8,812,171	B2	8/2014	Filev et al.
8,832,201	B2	9/2014	Wall
8,832,552	B2	9/2014	Arrasvuori et al.
8,839,327	B2	9/2014	Amento et al.
8,856,691	B2	10/2014	Geisner et al.
8,890,926	B2	11/2014	Tandon et al.
8,891,880	B2	11/2014	Oami et al.
8,892,999	B2	11/2014	Nims et al.
8,924,250	B2	12/2014	Bates et al.
8,963,926	B2	2/2015	Brown et al.
8,989,786	B2	3/2015	Feghali
9,086,776	B2	7/2015	Ye et al.
9,104,907	B2	8/2015	Whitehill et al.
9,105,014	B2	8/2015	Collet et al.
9,225,897	B1	12/2015	Sehn et al.
9,230,160	B1	1/2016	Kanter
9,241,184	B2	1/2016	Weerasinghe
9,251,170	B2	2/2016	Moore et al.
9,256,860	B2	2/2016	Herger et al.
9,276,886	B1	3/2016	Samaranayake
9,298,257	B2	3/2016	Hwang et al.
9,314,692	B2	4/2016	Konoplev et al.
9,330,483	B2	5/2016	Du et al.
9,357,174	B2	5/2016	Li et al.
9,361,510	B2	6/2016	Yao et al.
9,378,576	B2	6/2016	Bouaziz et al.
9,402,057	B2	7/2016	Kaytaz et al.
9,412,192	B2	8/2016	Mandel et al.
9,460,541	B2	10/2016	Li et al.
9,489,760	B2	11/2016	Li et al.
9,503,845	B2	11/2016	Vincent
9,508,197	B2	11/2016	Quinn et al.
9,532,364	B2	12/2016	Fujito
9,544,257	B2	1/2017	Ogundokun et al.
9,576,400	B2	2/2017	Van Os et al.
9,589,357	B2	3/2017	Li et al.
9,592,449	B2	3/2017	Barbalet et al.
9,648,376	B2	5/2017	Chang et al.
9,697,635	B2	7/2017	Quinn et al.
9,705,831	B2	7/2017	Spiegel
9,706,040	B2	7/2017	Kadirvel et al.
9,742,713	B2	8/2017	Spiegel et al.
9,744,466	B2	8/2017	Fujioka
9,746,990	B2	8/2017	Anderson et al.
9,749,270	B2	8/2017	Collet et al.
9,792,714	B2	10/2017	Li et al.
9,839,844	B2	12/2017	Dunstan et al.
9,883,838	B2	2/2018	Kaleal, III et al.
9,898,742	B2	2/2018	Higgins et al.
9,898,849	B2	2/2018	Du et al.
9,911,073	B1	3/2018	Spiegel et al.
9,936,165	B2	4/2018	Li et al.
9,959,037	B2	5/2018	Chaudhri et al.
9,965,801	B1	5/2018	Wishne et al.
9,980,100	B1	5/2018	Charlton et al.
9,990,373	B2	6/2018	Fortkort
10,039,988	B2	8/2018	Lobb et al.
10,097,492	B2	10/2018	Tsuda et al.
10,102,423	B2	10/2018	Shaburov et al.
10,109,315	B2	10/2018	Saban et al.
10,116,598	B2	10/2018	Tucker et al.
10,155,168	B2	12/2018	Blackstock et al.
10,158,589	B2	12/2018	Collet et al.
10,176,636	B1	1/2019	Neustein et al.
10,217,286	B1	2/2019	Angel et al.
10,242,477	B1	3/2019	Charlton et al.
10,242,503	B2	3/2019	McPhee et al.
10,262,250	B1	4/2019	Spiegel et al.
10,282,057	B1	5/2019	Binder et al.
10,284,508	B1	5/2019	Allen et al.
10,348,662	B2	7/2019	Baldwin et al.
10,362,219	B2	7/2019	Wilson et al.
10,402,689	B1	9/2019	Bogdanovych et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,432,559 B2 10/2019 Baldwin et al.
10,439,972 B1 10/2019 Spiegel et al.
10,454,857 B1 10/2019 Blackstock et al.
10,475,225 B2 11/2019 Park et al.
10,504,266 B2 12/2019 Blattner et al.
10,509,466 B1 12/2019 Miller et al.
10,514,876 B2 12/2019 Sehn
10,529,137 B1 1/2020 Black et al.
10,573,048 B2 2/2020 Ni et al.
10,579,869 B1 3/2020 Xiong et al.
10,607,411 B1 3/2020 Pezzino et al.
10,614,855 B2 4/2020 Huang
10,656,797 B1 5/2020 Alvi et al.
10,657,695 B2 5/2020 Chand et al.
10,657,701 B2 5/2020 Osman et al.
10,699,104 B2 6/2020 Xu et al.
10,748,347 B1 8/2020 Li et al.
10,762,174 B2 9/2020 Denton et al.
10,805,248 B2 10/2020 Luo et al.
10,835,828 B1 11/2020 Pather et al.
10,861,248 B2 12/2020 Scapel et al.
10,872,451 B2 12/2020 Sheth et al.
10,880,246 B2 12/2020 Baldwin et al.
10,895,964 B1 1/2021 Grantham et al.
10,896,534 B1 1/2021 Smith et al.
10,933,311 B2 3/2021 Brody et al.
10,938,758 B2 3/2021 Allen et al.
10,958,608 B1 3/2021 Allen et al.
10,962,809 B1 3/2021 Castañeda
10,964,082 B2 3/2021 Amitay et al.
10,979,752 B1 4/2021 Brody et al.
10,984,575 B2 4/2021 Assouline et al.
10,992,619 B2 4/2021 Antmen et al.
10,996,846 B2 5/2021 Robertson et al.
10,997,787 B2 5/2021 Ge et al.
11,010,022 B2 5/2021 Alvi et al.
11,012,390 B1 5/2021 Al Majid et al.
11,030,454 B1 6/2021 Xiong et al.
11,030,789 B2 6/2021 Chand et al.
11,030,807 B2 6/2021 Watts et al.
11,036,368 B1 6/2021 Al Majid et al.
11,036,781 B1 6/2021 Baril et al.
11,062,498 B1 7/2021 Voss et al.
11,063,891 B2 7/2021 Voss
11,069,103 B1 7/2021 Blackstock et al.
11,080,917 B2 8/2021 Monroy-hernández et al.
11,087,728 B1 8/2021 Canberk et al.
11,092,998 B1 8/2021 Castañeda et al.
11,106,342 B1 8/2021 Al Majid et al.
11,126,206 B2 9/2021 Meisenholder et al.
11,128,586 B2 9/2021 Al Majid et al.
11,143,867 B2 10/2021 Rodriguez, II
11,169,600 B1 11/2021 Canberk et al.
11,182,615 B2 11/2021 Lin et al.
11,188,190 B2 11/2021 Blackstock et al.
11,189,070 B2 11/2021 Jahangiri et al.
11,199,957 B1 12/2021 Alvi et al.
11,202,017 B2 12/2021 Holzer et al.
11,218,433 B2 1/2022 Baldwin et al.
11,227,626 B1 1/2022 Krishnan Gorumkonda et al.
11,229,849 B2 1/2022 Blackstock et al.
11,245,658 B2 2/2022 Grantham et al.
11,249,614 B2 2/2022 Brody
11,263,254 B2 3/2022 Baril et al.
11,263,821 B1 3/2022 Singh et al.
11,270,491 B2 3/2022 Monroy-Hernández et al.
11,284,144 B2 3/2022 Kotsopoulos et al.
11,307,747 B2 4/2022 Dancie et al.
11,393,133 B2 7/2022 El Kaliouby et al.
11,531,402 B1 12/2022 Stolzenberg
11,546,505 B2 1/2023 Canberk
11,636,662 B2 4/2023 Berger et al.
11,651,572 B2 5/2023 Berger et al.
11,670,059 B2 6/2023 Berger et al.
11,673,054 B2 6/2023 Berger et al.
11,734,866 B2 8/2023 Berger et al.
11,790,531 B2 10/2023 Dudovitch et al.
11,900,506 B2 2/2024 Berger et al.
11,983,826 B2 5/2024 Berger et al.
12,056,832 B2 8/2024 Berger et al.
12,100,156 B2 9/2024 Dudovitch et al.
12,148,108 B2 11/2024 Berger et al.
12,198,664 B2 1/2025 Berger et al.
2002/0067362 A1 6/2002 Agostino Nocera et al.
2002/0169644 A1 11/2002 Greene
2005/0162419 A1 7/2005 Kim et al.
2005/0206610 A1 9/2005 Cordelli
2006/0197779 A1 9/2006 Gossman et al.
2006/0294465 A1 12/2006 Ronen et al.
2007/0113181 A1 5/2007 Blattner et al.
2007/0168863 A1 7/2007 Blattner et al.
2007/0176921 A1 8/2007 Iwasaki et al.
2008/0158222 A1 7/2008 Li et al.
2009/0012788 A1 1/2009 Gilbert et al.
2009/0016617 A1 1/2009 Bregman-Amitai et al.
2009/0055484 A1 2/2009 Vuong et al.
2009/0070688 A1 3/2009 Gyorfi et al.
2009/0099925 A1 4/2009 Mehta et al.
2009/0106672 A1 4/2009 Burstrom
2009/0158170 A1 6/2009 Narayanan et al.
2009/0177976 A1 7/2009 Bokor et al.
2009/0202114 A1 8/2009 Morin et al.
2009/0231356 A1 9/2009 Barnes et al.
2009/0265604 A1 10/2009 Howard et al.
2009/0300525 A1 12/2009 Jolliff et al.
2009/0303984 A1 12/2009 Clark et al.
2010/0011422 A1 1/2010 Mason et al.
2010/0023885 A1 1/2010 Reville et al.
2010/0115426 A1 5/2010 Liu et al.
2010/0162149 A1 6/2010 Sheleheda et al.
2010/0173709 A1 7/2010 Horovitz et al.
2010/0203968 A1 8/2010 Gill et al.
2010/0227682 A1 9/2010 Reville et al.
2011/0007174 A1 1/2011 Bacivarov et al.
2011/0093780 A1 4/2011 Dunn
2011/0115798 A1 5/2011 Nayar et al.
2011/0148864 A1 6/2011 Lee et al.
2011/0239136 A1 9/2011 Goldman et al.
2011/0294564 A1 12/2011 Michelstein et al.
2011/0298897 A1 12/2011 Sareen et al.
2011/0301934 A1 12/2011 Tardif
2012/0113106 A1 5/2012 Choi et al.
2012/0124458 A1 5/2012 Cruzada
2012/0130717 A1 5/2012 Xu et al.
2013/0103760 A1 4/2013 Golding et al.
2013/0113830 A1* 5/2013 Suzuki G06T 19/20
345/634
2013/0147933 A1 6/2013 Kulas
2013/0185679 A1 7/2013 Fretwell et al.
2013/0201187 A1 8/2013 Tong et al.
2013/0249948 A1 9/2013 Reitan
2013/0257877 A1 10/2013 Davis
2013/0293530 A1 11/2013 Perez et al.
2014/0010449 A1 1/2014 Haaramo et al.
2014/0043329 A1 2/2014 Wang et al.
2014/0055554 A1 2/2014 Du et al.
2014/0125678 A1 5/2014 Wang et al.
2014/0129343 A1 5/2014 Finster et al.
2014/0171036 A1 6/2014 Simmons
2014/0176565 A1 6/2014 Adeyoola et al.
2014/0226000 A1 8/2014 Vilcovsky et al.
2014/0244447 A1 8/2014 Kim et al.
2015/0120293 A1 4/2015 Wohlert et al.
2015/0206349 A1 7/2015 Rosenthal et al.
2015/0287403 A1 10/2015 Holzer Zaslansky et al.
2015/0370320 A1 12/2015 Connor
2016/0055370 A1 2/2016 Garcia
2016/0127710 A1 5/2016 Saban et al.
2016/0134840 A1 5/2016 Mcculloch
2016/0234149 A1 8/2016 Tsuda et al.
2016/0343164 A1 11/2016 Urbach et al.
2017/0080346 A1 3/2017 Abbas
2017/0087473 A1 3/2017 Siegel et al.
2017/0091535 A1 3/2017 Yu et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2017/0091570 A1 3/2017 Rao et al.
2017/0113140 A1 4/2017 Blackstock et al.
2017/0118145 A1 4/2017 Aittoniemi et al.
2017/0123487 A1 5/2017 Hazra et al.
2017/0199855 A1 7/2017 Fishbeck
2017/0235848 A1 8/2017 Van Deusen et al.
2017/0256084 A1 9/2017 Iglehart et al.
2017/0272651 A1 9/2017 Mathy et al.
2017/0277684 A1 9/2017 Dharmarajan Mary
2017/0277685 A1 9/2017 Takumi
2017/0310934 A1 10/2017 Du et al.
2017/0312634 A1 11/2017 Ledoux et al.
2017/0337693 A1 11/2017 Baruch
2017/0351910 A1 12/2017 Elwazer et al.
2018/0032818 A1 2/2018 Abraham et al.
2018/0047200 A1 2/2018 O'hara et al.
2018/0089883 A1 3/2018 De Goes et al.
2018/0101510 A1 4/2018 Taylor
2018/0113587 A1 4/2018 Allen et al.
2018/0115503 A1 4/2018 Baldwin et al.
2018/0129870 A1 5/2018 Bacivarov et al.
2018/0158370 A1 6/2018 Pryor
2018/0182171 A1 6/2018 Lipner
2018/0198743 A1 7/2018 Blackstock et al.
2018/0240280 A1 8/2018 Chen et al.
2018/0286083 A1 10/2018 Jeon et al.
2018/0315076 A1 11/2018 Andreou
2018/0315133 A1 11/2018 Brody et al.
2018/0315134 A1 11/2018 Amitay et al.
2018/0330152 A1 11/2018 Mittelstaedt et al.
2019/0001223 A1 1/2019 Blackstock et al.
2019/0035153 A1 1/2019 Dange
2019/0050427 A1 2/2019 Wiesel et al.
2019/0057616 A1 2/2019 Cohen et al.
2019/0080457 A1 3/2019 Shukla et al.
2019/0097958 A1 3/2019 Collet et al.
2019/0116322 A1 4/2019 Holzer et al.
2019/0121522 A1 4/2019 Davis et al.
2019/0130652 A1 5/2019 Yu et al.
2019/0138796 A1 5/2019 Nakata et al.
2019/0188920 A1 6/2019 Mcphee et al.
2019/0196698 A1 6/2019 Cohen et al.
2019/0205627 A1 7/2019 De La Torre et al.
2019/0280997 A1 9/2019 Baldwin et al.
2019/0325279 A1 10/2019 Wang et al.
2019/0325631 A1 10/2019 Cao et al.
2019/0347803 A1 11/2019 Lim et al.
2019/0371028 A1 12/2019 Harrises et al.
2020/0005544 A1* 1/2020 Kim G06T 13/40
2020/0019364 A1 1/2020 Pond
2020/0021752 A1 1/2020 Holzer et al.
2020/0066044 A1 2/2020 Stahl et al.
2020/0066052 A1 2/2020 Antonsen et al.
2020/0117938 A1 4/2020 Greenberger et al.
2020/0151807 A1 5/2020 Zhou et al.
2020/0219295 A1 7/2020 El Kaliouby et al.
2020/0228729 A1 7/2020 Okuno et al.
2020/0302175 A1 9/2020 Mason
2020/0306637 A1 10/2020 Baldwin et al.
2020/0344407 A1 10/2020 Ono et al.
2020/0372127 A1 11/2020 Denton et al.
2020/0410575 A1 12/2020 Grantham et al.
2021/0011612 A1 1/2021 Dancie et al.
2021/0027100 A1 1/2021 Bogdanovych et al.
2021/0074016 A1 3/2021 Li et al.
2021/0074047 A1 3/2021 Sheth et al.
2021/0089179 A1 3/2021 Grantham et al.
2021/0104087 A1 4/2021 Smith et al.
2021/0118237 A1* 4/2021 Hou G06V 10/751
2021/0166732 A1 6/2021 Shaburova et al.
2021/0168108 A1 6/2021 Antmen et al.
2021/0170270 A1 6/2021 Brody et al.
2021/0174034 A1 6/2021 Retek et al.
2021/0174592 A1 6/2021 Zhang et al.
2021/0192823 A1 6/2021 Amitay et al.
2021/0209825 A1 7/2021 Assouline et al.
2021/0225058 A1 7/2021 Chand et al.
2021/0240315 A1 8/2021 Alvi et al.
2021/0241529 A1 8/2021 Cowburn et al.
2021/0243482 A1 8/2021 Baril et al.
2021/0243503 A1 8/2021 Kotsopoulos et al.
2021/0266277 A1 8/2021 Allen et al.
2021/0281897 A1 9/2021 Brody et al.
2021/0285774 A1 9/2021 Collins et al.
2021/0287274 A1 9/2021 Nguyen et al.
2021/0303075 A1 9/2021 Cowburn et al.
2021/0303077 A1 9/2021 Anvaripour et al.
2021/0303140 A1 9/2021 Mourkogiannis
2021/0303855 A1 9/2021 Anvaripour et al.
2021/0306290 A1 9/2021 Voss
2021/0306451 A1 9/2021 Heikkinen et al.
2021/0375320 A1 12/2021 Wong et al.
2021/0382564 A1 12/2021 Blachly et al.
2021/0385180 A1 12/2021 Al Majid et al.
2021/0397000 A1 12/2021 Rodriguez, II
2021/0398095 A1 12/2021 Mallett et al.
2021/0405761 A1 12/2021 Canberk
2021/0405831 A1 12/2021 Mourkogiannis et al.
2021/0409535 A1 12/2021 Mourkogiannis et al.
2022/0012929 A1 1/2022 Blackstock et al.
2022/0083768 A1 3/2022 Johnson et al.
2022/0101417 A1 3/2022 Boscolo et al.
2022/0108470 A1 4/2022 Brown et al.
2022/0188539 A1 6/2022 Chan et al.
2022/0206588 A1 6/2022 Canberk et al.
2022/0237799 A1 7/2022 Price et al.
2022/0270261 A1 8/2022 Dudovitch et al.
2022/0270265 A1 8/2022 Dudovitch et al.
2022/0300730 A1 9/2022 Eirinberg et al.
2022/0300731 A1 9/2022 Eirinberg et al.
2022/0326781 A1 10/2022 Hwang et al.
2022/0327709 A1 10/2022 Dudovitch et al.
2022/0334649 A1 10/2022 Hwang et al.
2022/0374968 A1 11/2022 Berger et al.
2022/0377257 A1 11/2022 Wilson et al.
2023/0065031 A1 3/2023 Berger et al.
2023/0066179 A1 3/2023 Berger et al.
2023/0070631 A1 3/2023 Berger et al.
2023/0075256 A1 3/2023 Berger et al.
2023/0078483 A1 3/2023 Berger et al.
2023/0085099 A1 3/2023 Ouyang et al.
2023/0100509 A1 3/2023 Berger et al.
2023/0103125 A1 3/2023 Berger et al.
2023/0116652 A1 4/2023 Berger et al.
2023/0154129 A1 5/2023 Berger et al.
2023/0169739 A1 6/2023 Berger et al.
2023/0281936 A1 9/2023 Berger et al.
2023/0343007 A1 10/2023 Berger et al.
2023/0419497 A1 12/2023 Dudovitch et al.
2024/0029322 A1 1/2024 Berger et al.
2024/0161423 A1 5/2024 Sholl et al.
2024/0338913 A1 10/2024 Berger et al.
2025/0095618 A1 3/2025 Berger et al.

FOREIGN PATENT DOCUMENTS

CN 109863532 A 6/2019
CN 110168478 A 8/2019
CN 110310319 A 10/2019
CN 110096156 B 6/2021
CN 116917938 A 10/2023
CN 117136381 A 11/2023
CN 117157667 A 12/2023
CN 117897734 A 4/2024
CN 117940962 A 4/2024
CN 117957043 A 4/2024
EP 2184092 A2 5/2010
EP 3155560 B1 5/2020
EP 3707693 A1 9/2020
EP 3772040 A1 2/2021
GB 2482729 A 2/2012
JP 2001230801 A 8/2001
JP 5497931 B2 3/2014
JP 5722603 B2 4/2015

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

KR	101445263	B1	9/2014
KR	20190097815	A	8/2019
KR	102181648	B1	11/2020
KR	20220158824	A	12/2022
KR	20240056547	A	4/2024
WO	WO-2003094072	A1	11/2003
WO	WO-2004095308	A1	11/2004
WO	WO-2006107182	A1	10/2006
WO	WO-2007134402	A1	11/2007
WO	WO-2012139276	A1	10/2012
WO	WO-2013027893	A1	2/2013
WO	WO-2013152454	A1	10/2013
WO	WO-2013166588	A1	11/2013
WO	WO-2014031899	A1	2/2014
WO	WO-2014194439	A1	12/2014
WO	WO-2016054164	A1	4/2016
WO	WO-2016090605	A1	6/2016
WO	WO-2016168591	A1	10/2016
WO	WO-2017008427	A1	1/2017
WO	WO-2017203262	A2	11/2017
WO	WO-2018075053	A1	4/2018
WO	WO-2018081013	A1	5/2018
WO	WO-2018102562	A1	6/2018
WO	WO-2018129531	A1	7/2018
WO	WO-2019089613	A1	5/2019
WO	WO-2019094618	A1	5/2019
WO	WO-2019162842	A1	8/2019
WO	WO-2020163592	A1	8/2020
WO	WO-2021174139	A1	9/2021
WO	WO-2022005687	A1	1/2022
WO	WO-2022005693	A1	1/2022
WO	WO-2022060549	A2	3/2022
WO	WO-2022066578	A1	3/2022
WO	WO-2022125167	A1	6/2022
WO	WO-2022132381	A1	6/2022
WO	WO-2022146678	A1	7/2022
WO	WO-2022182660	A1	9/2022
WO	WO-2022183175	A1	9/2022
WO	WO-2022198182	A1	9/2022
WO	WO-2022216784	A1	10/2022
WO	WO-2022221243	A2	10/2022
WO	WO-2022221243	A3	10/2022
WO	WO-2022225761	A1	10/2022
WO	WO-2022245765	A1	11/2022
WO	WO-2023034832	A1	3/2023
WO	WO-2023039183	A1	3/2023
WO	WO-2023039390	A1	3/2023
WO	WO-2023055825	A1	4/2023
WO	WO-2023055842	A1	4/2023

OTHER PUBLICATIONS

"U.S. Appl. No. 17/301,690, Response filed Jan. 2, 2024 to Restriction Requirement mailed Nov. 21, 2023", 3 pgs.
"U.S. Appl. No. 17/301,690, Restriction Requirement mailed Nov. 21, 2023", 7 pgs.
"U.S. Appl. No. 17/446,691, Corrected Notice of Allowability mailed Feb. 15, 2023", 2 pgs.
"U.S. Appl. No. 17/446,691, Non Final Office Action mailed Jun. 20, 2022", 24 pgs.
"U.S. Appl. No. 17/446,691, Notice of Allowance mailed Jan. 25, 2023", 5 pgs.
"U.S. Appl. No. 17/446,691, Notice of Allowance mailed Oct. 3, 2022", 8 pgs.
"U.S. Appl. No. 17/446,691, Response filed Sep. 19, 2022 to Non Final Office Action mailed Jun. 20, 2022", 10 pgs.
"U.S. Appl. No. 17/447,509, Notice of Allowance mailed Apr. 3, 2023", 8 pgs.
"U.S. Appl. No. 17/468,379, Final Office Action mailed Oct. 24, 2022", 7 pgs.
"U.S. Appl. No. 17/468,379, Non Final Office Action mailed Jul. 6, 2022", 6 pgs.
"U.S. Appl. No. 17/468,379, Notice of Allowance mailed Jan. 10, 2023", 5 pgs.
"U.S. Appl. No. 17/468,379, Notice of Allowance mailed Apr. 26, 2023", 7 pgs.
"U.S. Appl. No. 17/468,379, Response filed Oct. 4, 2022 to Non Final Office Action mailed Jul. 6, 2022", 11 pgs.
"U.S. Appl. No. 17/468,379, Response filed Dec. 13, 2022 to Final Office Action mailed Oct. 24, 2022", 10 pgs.
"U.S. Appl. No. 17/470,231, Advisory Action mailed May 15, 2023", 3 pgs.
"U.S. Appl. No. 17/470,231, Corrected Notice of Allowability mailed Dec. 27, 2023", 3 pgs.
"U.S. Appl. No. 17/470,231, Final Office Action mailed Mar. 1, 2023", 27 pgs.
"U.S. Appl. No. 17/470,231, Non Final Office Action mailed Oct. 13, 2022", 22 pgs.
"U.S. Appl. No. 17/470,231, Notice of Allowance mailed Jul. 7, 2023", 8 pgs.
"U.S. Appl. No. 17/470,231, Response filed Jan. 10, 2023 to Non Final Office Action mailed Oct. 13, 2022", 9 pgs.
"U.S. Appl. No. 17/470,231, Response filed May 1, 2023 to Final Office Action mailed Mar. 1, 2023", 11 pgs.
"U.S. Appl. No. 17/490,167, Advisory Action mailed Jul. 17, 2023", 3 pgs.
"U.S. Appl. No. 17/490,167, Final Office Action mailed May 10, 2023", 18 pgs.
"U.S. Appl. No. 17/490,167, Final Office Action mailed Nov. 7, 2023", 17 pgs.
"U.S. Appl. No. 17/490,167, Non Final Office Action mailed Jan. 19, 2023", 13 pgs.
"U.S. Appl. No. 17/490,167, Non Final Office Action mailed Aug. 30, 2023", 16 pgs.
"U.S. Appl. No. 17/490,167, Notice of Allowance mailed Jan. 11, 2024", 9 pgs.
"U.S. Appl. No. 17/490,167, Response filed Apr. 4, 2023 to Non Final Office Action mailed Jan. 19, 2023", 10 pgs.
"U.S. Appl. No. 17/490,167, Response filed Jun. 28, 2023 to Final Office Action mailed May 10, 2023", 13 pgs.
"U.S. Appl. No. 17/490,167, Response filed Oct. 5, 2023 to Non Final Office Action mailed Aug. 30, 2023", 10 pgs.
"U.S. Appl. No. 17/490,167, Response filed Dec. 14, 2023 to Final Office Action mailed Nov. 7, 2023", 10 pgs.
"U.S. Appl. No. 17/490,209, Notice of Allowance mailed Mar. 10, 2023", 9 pgs.
"U.S. Appl. No. 17/490,209, Notice of Allowance mailed Jul. 21, 2022", 10 pgs.
"U.S. Appl. No. 17/490,209, Notice of Allowance mailed Nov. 23, 2022", 9 pgs.
"U.S. Appl. No. 17/498,475, Corrected Notice of Allowability mailed Apr. 5, 2023", 2 pgs.
"U.S. Appl. No. 17/498,475, Non Final Office Action mailed Aug. 5, 2022", 25 pgs.
"U.S. Appl. No. 17/498,475, Notice of Allowance mailed Mar. 8, 2023", 5 pgs.
"U.S. Appl. No. 17/498,475, Notice of Allowance mailed Nov. 16, 2022", 11 pgs.
"U.S. Appl. No. 17/498,475, Response filed Oct. 27, 2022 to Non Final Office Action mailed Aug. 5, 2022", 9 pgs.
"U.S. Appl. No. 18/138,269, Non Final Office Action mailed Dec. 21, 2023", 9 pgs.
"U.S. Appl. No. 18/138,269, Response filed Feb. 21, 2024 to Non Final Office Action mailed Dec. 21, 2023", 8 pgs.
"Application Serial No. PCT/US2022/024364, Invitation to Pay Additional Fees mailed Sep. 15, 2022", 12 pgs.
"Curiscope Virtuali-Tee app", as evidenced by a YouTube video, Curiscope's VirtualiTee: Wearable tech you learn with—YouTube, [Online] Retrieved from the Internet: <URL: https://www.youtube.com/watch?v=QOHfdqgwFU>, (Mar. 1, 2016), 1 pg.
"International Application Serial No. PCT/US2022/024364, International Preliminary Report on Patentability mailed Oct. 26, 2023", 13 pgs.
"International Application Serial No. PCT/US2022/024364, International Search Report mailed Nov. 8, 2022", 7 pgs.

(56) **References Cited**

OTHER PUBLICATIONS

"International Application Serial No. PCT/US2022/024364, Written Opinion mailed Nov. 8, 2022", 11 pgs.
"International Application Serial No. PCT/US2022/043094, International Search Report mailed Jan. 4, 2023", 5 pgs.
"International Application Serial No. PCT/US2022/043094, Written Opinion mailed Jan. 4, 2023", 7 pgs.
"International Application Serial No. PCT/US2022/045075, International Search Report mailed Jan. 26, 2023", 4 pgs.
"International Application Serial No. PCT/US2022/045075, Written Opinion mailed Jan. 26, 2023", 6 pgs.
"International Application Serial No. PCT/US2022/045099, International Search Report mailed Feb. 2, 2023", 5 pgs.
"International Application Serial No. PCT/US2022/045099, Written Opinion mailed Feb. 2, 2023", 9 pgs.
"International Application Serial No. PCT/US2022/075707, International Search Report mailed Dec. 20, 2022", 4 pgs.
"International Application Serial No. PCT/US2022/075707, Written Opinion mailed Dec. 20, 2022", 5 pgs.
"International Application Serial No. PCT/US2022/076006, International Search Report mailed Dec. 23, 2022", 4 pgs.
"International Application Serial No. PCT/US2022/076006, Written Opinion mailed Dec. 23, 2022", 4 pgs.
Aiyu, Cui, et al., "Dressing in Order: Recurrent Person Image Generation for Pose Transfer, Virtual Try-on and Outfit Editing", arxiv.org, Cornell University. Library, 201 Olin Library Cornell University Ithaca, NY 14853, (Aug. 31, 2021), 19 pgs.
Bradley, Derek, et al., "Augmented Reality on Cloth with Realistic Illumination", Machine Vision and Applications 20, (2009), 85-92.
Hashmi, Mohammad Farukh, et al., "FashionFit: Analysis of Mapping 3D Pose and Neural Body Fit for Custom Virtual Try-On", IEEE Access, IEEE, USA, vol. 8, (May 11, 2020), 91603-91615.
Hilsmann, et al., "Tracking and Retexturing Cloth for Real-Time Virtual Clothing Applications", Springer-Verlag Berlin Heidelberg 2009, (2009), 12 pgs.
Hobley, Michael A, et al., "Say Yes to the Dress: Shape and Style Transfer Using Conditional GAN", Advances in Databases and Information Systems; [Lecture Notes in Computer Science; Lect. Notes Computer], Springer International Publishing, Cham, (May 29, 2019), 135-149.
Hobley, Michael, et al., "Say Yes to the Dress: Shape and Style Transfer Using Conditional GANs", Advances in Databases and Information Systems; [Lecture Notes in Computer Science; Lect. Notes Computer], Springer International Publishing, Cham, Page(S), (May 29, 2019), 15 pgs.
Kanduri, et al., "A Modular Approach for Cloth Modelling in Virtual Fitting Room", ICIIS, (2016), 106-111.
Liu, Li, et al., "A data-driven editing framework for automatic 3D garment modeling", Multimed Tools Appl 76, 2017, (2017), 12597-12626.
Liu, Si, et al., "Fashion Parsing With Video Context", IEEE Transactions On Multimedia, IEEE, USA, vol. 17, No. 8, (Aug. 1, 2015), 1347-1358.
Martinsson, et al., "Semantic Segmentation of Fashion Images Using Feature Pyramid Networks", IEEE, (2019), 4 pgs.
Pons-Moll, et al., "Clothing Capture and Retargeting", ACM Transactions on Graphics, vol. 36, No. 4, (2017), 15 pgs.
Raj, Amit, et al., "SwapNet: Image Based Garment Transfer", Pattern Recognition: 5th Asian Conference, ACPR 2019, Auckland, NZ, Nov. 26-29, 2019, Revised Selected Papers, Part II; [Lecture Notes in Computer Science], Springer, Cham, (Oct. 6, 2018), 679-695.
Wang, et al., "Estimation of multiple directional light sources for synthesis of augmented reality images", Elsevier Science, Graphical Models 65, (Dec. 31, 2003), 185-205.
Xueping, Su, et al., "Personalized Clothing Recommendation Based on User Emotional Analysis", Discrete Dynamics in Nature and Society, vol. 2020, [Online]. Retrieved from the Internet: <URL:http://downloads.hindawi.com/journals/ddns/2020/7954393.XML>, (Mar. 5, 2020), 8 pgs.
Yang, et al., "Recognize the Silhouette Attributes of T-shirts based on Mask R-CNN", IEEE, (Apr. 2021), 5 pgs.
Yoon, Shin Jae, et al., "Neural 3D Clothes Retargeting from a Single Image", Cornell University Library, 201 Olin Library Cornell University Ithaca, NY 14853, (Jan. 29, 2021), 20 pgs.
Zhaoqi, Su, et al., "MulayCap: Multi-layer Human Performance Capture Using a Monocular Video Camera", arxiv.org, Cornell University Library, 201 Olin Library Cornell University Ithaca, NY 14853, (Oct. 1, 2020), 18 pgs.
U.S. Appl. No. 17/301,690, filed Apr. 12, 2021, Garment Segmentation.
U.S. Appl. No. 17/446,757, filed Sep. 2, 2021, Interactive Fashion With Music AR.
U.S. Appl. No. 17/446,691 U.S. Pat. No. 11,670,059, filed Sep. 1, 2021, Controlling Interactive Fashion Based on Body Gestures.
U.S. Appl. No. 18/138,269, filed Apr. 24, 2023, Controlling Interactive Fashion Based on Body Gestures.
U.S. Appl. No. 17/470,231 U.S. Pat. No. 11,900,506, filed Sep. 9, 2021, Controlling Interactive Fashion Based on Facial Expressions.
U.S. Appl. No. 18/376,161, filed Oct. 3, 2023, Controlling Interactive Fashion Based on Facial Expressions.
U.S. Appl. No. 17/447,509 U.S. Pat. No. 11,734,866, filed Sep. 13, 2021, Controlling Interactive Fashion Based on Voice.
U.S. Appl. No. 18/215,465, filed Jun. 28, 2023, Controlling Interactive Fashion Based on Voice.
U.S. Appl. No. 17/490,167, filed Sep. 30, 2021, 3D Upper Garment Tracking.
U.S. Appl. No. 17/498,475 U.S. Pat. No. 11,651,572, filed Oct. 11, 2021, Light and Rendering of Garments.
U.S. Appl. No. 18/101,673, filed Jan. 26, 2023, Light and Rendering of Garments.
U.S. Appl. No. 17/490,209 U.S. Pat. No. 11,636,662, filed Sep. 30, 2021, Body Normal Network Light and Rendering Control.
U.S. Appl. No. 18/098,799, filed Jan. 19, 2023, Body Normal Network Light and Rendering Control.
"U.S. Appl. No. 17/249,239, Final Office Action mailed Mar. 31, 2023", 27 pgs.
"U.S. Appl. No. 17/249,239, Non Final Office Action mailed Oct. 5, 2022", 25 pgs.
"U.S. Appl. No. 17/249,239, Notice of Allowance mailed Jun. 7, 2023", 9 pgs.
"U.S. Appl. No. 17/249,239, Response filed May 17, 2023 to Final Office Action mailed Mar. 31, 2023", 10 pgs.
"U.S. Appl. No. 17/249,239, Response filed Dec. 20, 2022 to Non Final Office Action mailed Oct. 5, 2022", 10 pgs.
"U.S. Appl. No. 17/249,241, Non Final Office Action mailed Feb. 16, 2024", 21 pgs.
"U.S. Appl. No. 17/301,690, Notice of Allowance mailed May 22, 2024", 10 pgs.
"U.S. Appl. No. 17/301,690, Response filed Apr. 3, 2024 to Non Final Office Action mailed Jan. 19, 2024", 11 pgs.
"U.S. Appl. No. 17/446,757, Examiner Interview Summary mailed Aug. 1, 2024", 2 pgs.
"U.S. Appl. No. 17/446,757, Non Final Office Action mailed May 20, 2024", 10 pgs.
"U.S. Appl. No. 17/446,757, Notice of Allowance mailed Sep. 3, 2024", 11 pgs.
"U.S. Appl. No. 17/446,757, Response filed Aug. 20, 2024 to Non Final Office Action mailed May 20, 2024", 11 pgs.
"U.S. Appl. No. 18/101,673, Corrected Notice of Allowability mailed Oct. 21, 2024", 2 pgs.
"U.S. Appl. No. 18/101,673, Non Final Office Action mailed Mar. 15, 2024", 11 pgs.
"U.S. Appl. No. 18/101,673, Notice of Allowance mailed Jul. 5, 2024", 9 pgs.
"U.S. Appl. No. 18/101,673, Response filed Jun. 5, 2024 to Non Final Office Action mailed Mar. 15, 2024", 8 pgs.
"U.S. Appl. No. 18/138,269, Corrected Notice of Allowability mailed Jul. 8, 2024", 2 pgs.
"U.S. Appl. No. 18/138,269, Notice of Allowance mailed Mar. 20, 2024", 8 pgs.
"U.S. Appl. No. 18/215,465, Examiner Interview Summary mailed Oct. 7, 2024", 2 pgs.

(56) **References Cited**

OTHER PUBLICATIONS

"U.S. Appl. No. 18/215,465, Final Office Action mailed Dec. 16, 2024", 11 pgs.
"U.S. Appl. No. 18/215,465, Non Final Office Action mailed Jul. 5, 2024", 11 pgs.
"U.S. Appl. No. 18/215,465, Response filed Sep. 10, 2024 to Non Final Office Action mailed Jul. 5, 2024", 9 pgs.
"U.S. Appl. No. 18/376,161, Examiner Interview Summary mailed Oct. 7, 2024", 2 pgs.
"U.S. Appl. No. 18/376,161, Final Office Action mailed Dec. 23, 2024", 43 pgs.
"U.S. Appl. No. 18/376,161, Non Final Office Action mailed Aug. 9, 2024", 41 pgs.
"U.S. Appl. No. 18/376,161, Response filed Oct. 9, 2024 to Non Final Office Action mailed Aug. 9, 2024", 11 pgs.
"Bitmoji", Snapchat Support, [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20190503063620/https://support.snapchat.com/en-US/a/bitmoji>, (captured May 3, 2019), 2 pgs.
"Bitmoji Chrome Extension", [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20200919024925/https://support.bimoji.com/hc/en-us/articles/360001494066>, (Sep. 19, 2020), 5 pgs.
"Bitmoji Customize text", [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20210225200456/https://support.bitmoji.com/hc/en-us/articles/360034632291-Customize-Text-on-Bitmoji-Stickers>, (captured Feb. 25, 2021), 3 pgs.
"Bitmoji Family", Snapchat Support, [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20190503063620/https://support.snapchat.com/en-US/article/bitmoji-family>, (captured May 3, 2019), 4 pgs.
"Clone Squad AR effect", Superfan Studio, <https://www.youtube.com/watch?v=IWRCD3BrRYg>, (May 5, 2020), 1 pg.
"How to get invisible tiktok filter/effect use the invisible filter on tiktok", How To Digital, <https: //www.youtube.com/watch?v=qcYEwiLceY>, (Jan. 21, 2020), 1 pg.
"Image inpainting", Yokoyalab, <https://www.youtube.com/watch?v=2ve-zK-bYRo&t=20s>, (Feb. 22, 2012), 1 pg.
"Instant Comics Starring You & Your Friends", Bitstrips Inc, [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20150206000940/http://company.bitstrips.com/bitstrips-app.html>, (captured Feb. 6, 2015), 3 pgs.
"International Application Serial No. PCT/US2022/017322, International Preliminary Report on Patentability mailed Sep. 7, 2023", 13 pgs.
"International Application Serial No. PCT/US2022/017322, International Search Report mailed Jul. 28, 2022", 7 pgs.
"International Application Serial No. PCT/US2022/017322, Invitation to Pay Additional Fees mailed Jun. 7, 2022", 13 pgs.
"International Application Serial No. PCT/US2022/017322, Written Opinion mailed Jul. 28, 2022", 11 pgs.
"International Application Serial No. PCT/US2022/043094, International Preliminary Report on Patentability mailed Mar. 21, 2024", 9 pgs.
"International Application Serial No. PCT/US2022/045075, International Preliminary Report on Patentability mailed Apr. 11, 2024", 9 pgs.
"International Application Serial No. PCT/US2022/045099, International Preliminary Report on Patentability mailed Apr. 11, 2024", 11 pgs.
"International Application Serial No. PCT/US2022/070753, International Preliminary Report on Patentability mailed Sep. 7, 2023", 8 pgs.
"International Application Serial No. PCT/US2022/070753, International Search Report mailed May 27, 2022", 5 pgs.
"International Application Serial No. PCT/US2022/070753, Written Opinion mailed May 27, 2022", 6 pgs.
"International Application Serial No. PCT/US2022/075707, International Preliminary Report on Patentability mailed Mar. 14, 2024", 7 pgs.
"International Application Serial No. PCT/US2022/076006, International Preliminary Report on Patentability mailed Mar. 21, 2024", 6 pgs.
"Manage Your Bitmoji", Snapchat Support, [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20190503063620/https://support.snapchat.com/en-US/a/manage-bitmoji>, (captured May 3, 2019), 3 pgs.
"Your Own Personal Emoji", Bitstrips Inc, [Online] Retrieved from the Internet: <URL: https://web.archive.org/web/20150205232004/http://bitmoji.com/>, (captured Feb. 5, 2015), 3 pgs.
Carnahan, Daniel, "Snap is Offering Personalized Video Content Through Bitmoji TV", Business Insider, [Online] Retrieved from the Internet: <URL: https://www.businessinsider.com/snap-offers-personalized-video-content-through-bitmoji-tv-2019-12>, (2019), 10 pgs.
Chen, Xi, et al., "Real-time Human Segmentation using Pose Skeleton Map", 2019 Chinese Control Conference (CCC), Technical Committee on Control Theory, Chinese Association of Automation, (Jul. 27, 2019), 6 pgs.
Constine, Josh, "Snapchat launches Bitmoji merch and comic strips starring your avatar", TechCrunch, [Online] Retrieved from the Internet: <URL: https://techcrunch.com/2018/11/13/bitmoji-store/>, (Nov. 13, 2018), 16 pgs.
Constine, Josh, "Snapchat Launches Bitmoji TV: Zany 4-min Cartoons of Your Avatar", TechCrunch, [Online] Retrieved from the Internet: <URL: https://techcrunch.com/2020/01/30/bitmoji-tv/>, (Jan. 30, 2020), 13 pgs.
Fei-Fei, Li, et al., "One-Shot Learning of Object Categories", IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 28, No. 4, (Apr. 2006), 18 pgs.
Liu, Ting, et al., "3D Clothing Transfer in Virtual Fitting Based on UV Mapping", 11th International Congress on Image and Signal Processing, Biomedical Engineering and Informatics (CISP-BMEI), IEEE, (Oct. 13, 2018), 6 pgs.
MacMillan, Douglas, "Snapchat Buys Bitmoji App for More Than $100 Million", The Wallstreet Journal, [Online] Retrieved from the Internet: <URL: https://www.wsj.com/articles/snapchat-buys-bitmoji-app-for-more-than-100-million-1458876017>, (Mar. 25, 2016), 5 pgs.
Newton, Casey, "Your Snapchat friendships now have their own profiles—and merchandise", The Verge, [Online] Retrieved from the Internet: <URL: https://www.theverge.com/2018/11/13/18088772/snapchat-friendship-profiles-bitmoji-merchandise-comics>, (Nov. 13, 2018), 5 pgs.
Ong, Thuy, "Snapchat takes Bitmoji deluxe with hundreds of new customization options", The Verge, [Online] Retrieved from the Internet on Nov. 2, 2018: < URL: https://www.theverge.com/2018/1/30/16949402/bitmoji-deluxe-snapchat-customization>, (Jan. 30, 2018), 2 pgs.
Reign, Ashley, "How to Add My Friend's Bitmoji to My Snapchat", Women.com, [Online] Retrieved from the Internet: <URL: https://www.women.com/ashleyreign/lists/how-to-add-my-friends-bitmoji-to-my-snapchat>, (Jun. 30, 2017), 7 pgs.
Tumbokon, Karen, "Snapchat Update: How to Add Bitmoji to Customizable Geofilters", International Business Times, [Online] Retrieved from the Internet : <URL: https://www.ibtimes.com/snapchat-update-how-add-bitmoji-customizable-geofilters-2448152>, (Nov. 18, 2016), 6 pgs.
Wang, Lu, et al., "Crowd Counting and Segmentation in Visual Surveillance", 16th IEEE International Conference on Image Processing (ICIP), (Nov. 7-10, 2009), 4 pgs.
Zhao, Hengshuang, et al., "ICNet for Real-Time Semantic Segmentation on High-Resolution Images", Proceedings of the European Conference on Computer Vision (ECCV), (2018), 16 pgs.
"U.S. Appl. No. 18/098,799, Examiner Interview Summary mailed Mar. 25, 2025", 2 pgs.
"U.S. Appl. No. 18/098,799, Response filed Mar. 27, 2025 to Non Final Office Action mailed Feb. 13, 2025", 10 pgs.
"U.S. Appl. No. 18/215,465, Advisory Action mailed Mar. 3, 2025", 3 pgs.
"U.S. Appl. No. 18/215,465, Notice of Allowance mailed Mar. 31, 2025", 9 pgs.

(56) **References Cited**

OTHER PUBLICATIONS

"U.S. Appl. No. 18/376,161, Corrected Notice of Allowability mailed Apr. 1, 2025", 2 pgs.
"U.S. Appl. No. 18/376,161, Notice of Allowance mailed Mar. 19, 2025", 9 pgs.
"U.S. Appl. No. 18/098,799, Non Final Office Action mailed Feb. 13, 2025", 28 pgs.
"U.S. Appl. No. 18/215,465, Examiner Interview Summary mailed Jan. 27, 2025", 2 pgs.
"U.S. Appl. No. 18/215,465, Response filed Feb. 17, 2025 to Final Office Action mailed Dec. 16, 2024", 9 pgs.
"U.S. Appl. No. 18/376,161, Advisory Action mailed Feb. 14, 2025", 3 pgs.
"U.S. Appl. No. 18/376,161, Examiner Interview Summary mailed Jan. 27, 2025", 2 pgs.
"U.S. Appl. No. 18/376,161, Response filed Feb. 3, 2025 to Final Office Action mailed Dec. 23, 2024", 11 pgs.
"Korean Application Serial No. 10-2024-7011184, Notice of Preliminary Rejection mailed Jul. 10, 2025", W/English Translation, 11 pgs.
"Korean Application Serial No. 10-2024-7013344, Notice of Preliminary Rejection mailed Jun. 30, 2025", W/ English Translation, 14 pgs.
Pons-Moll, Gerard, et al., "ClothCap: Seamless 4D Clothing Capture and Retargeting", ACM Transactions on Graphics, vol. 36, No. 4, (2017), 15 pgs.

* cited by examiner

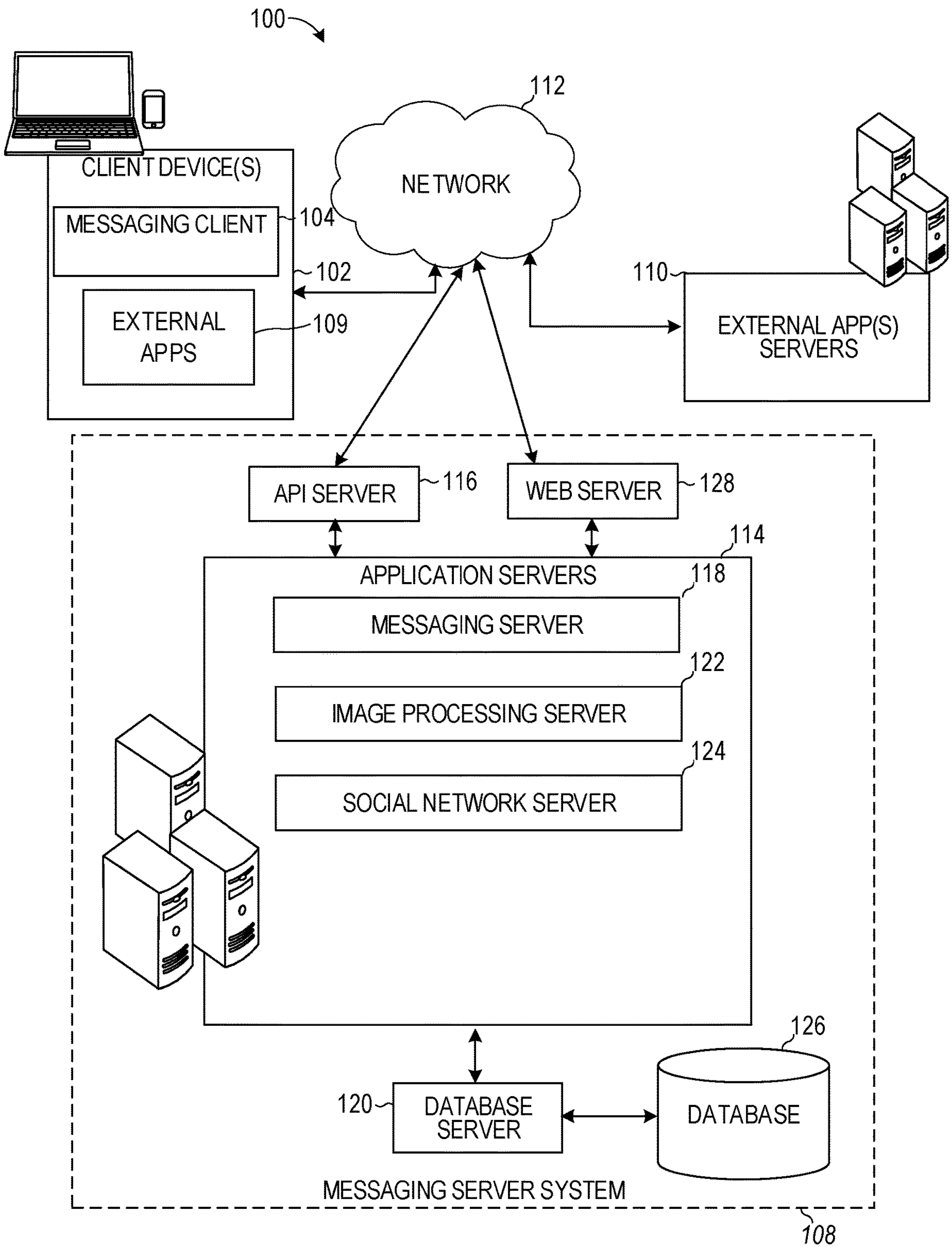
100
112
NETWORK
CLIENT DEVICE(S)
MESSAGING CLIENT
104
102
EXTERNAL APPS
109
110
EXTERNAL APP(S) SERVERS
API SERVER
116
WEB SERVER
128
114
APPLICATION SERVERS
118
MESSAGING SERVER
122
IMAGE PROCESSING SERVER
124
SOCIAL NETWORK SERVER
126
120
DATABASE SERVER
DATABASE
MESSAGING SERVER SYSTEM
108

FIG. 1

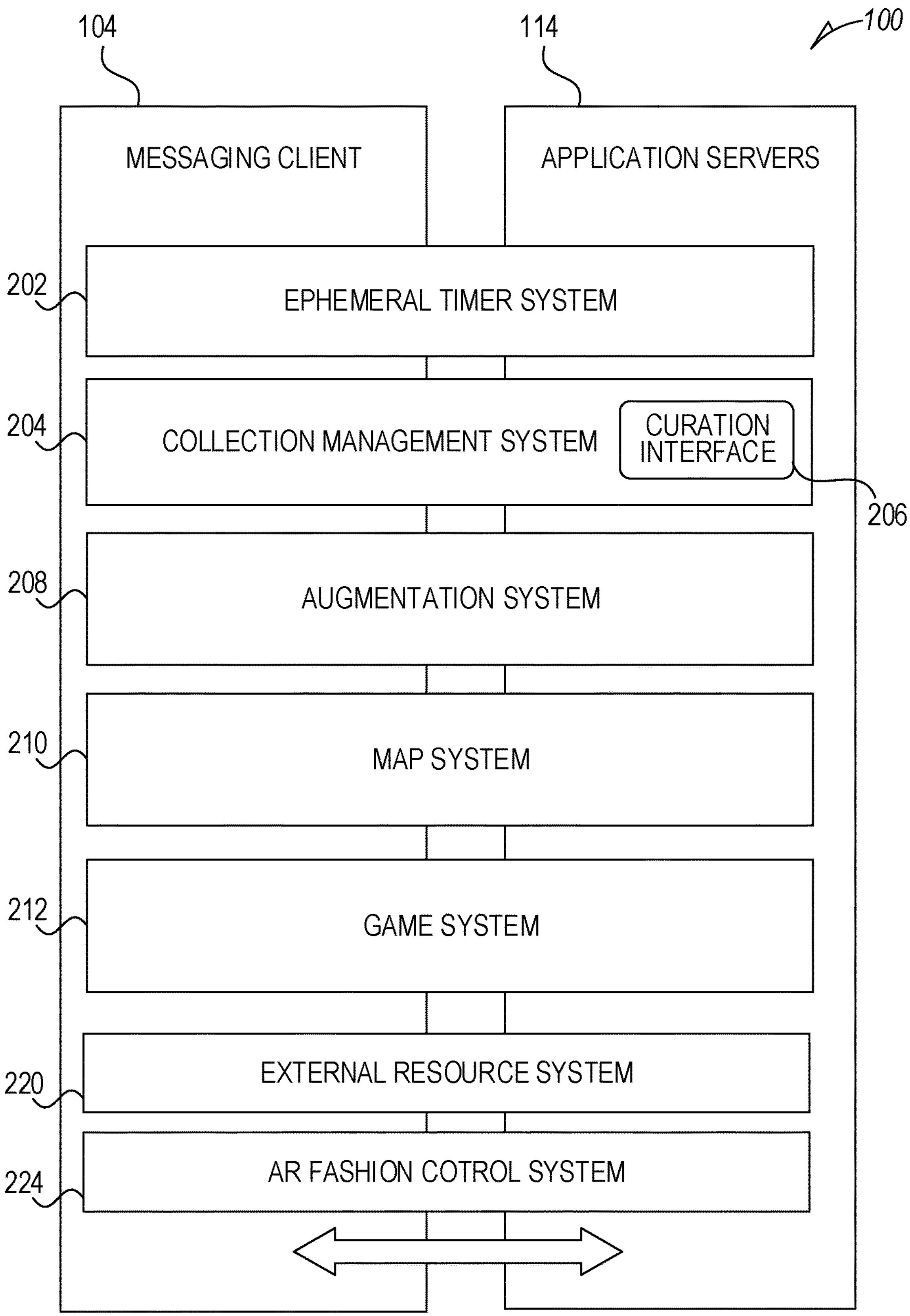
100
104
114
MESSAGING CLIENT
APPLICATION SERVERS
202
EPHEMERAL TIMER SYSTEM
204
COLLECTION MANAGEMENT SYSTEM
CURATION INTERFACE
206
208
AUGMENTATION SYSTEM
210
MAP SYSTEM
212
GAME SYSTEM
220
EXTERNAL RESOURCE SYSTEM
224
AR FASHION COTROL SYSTEM

FIG. 2

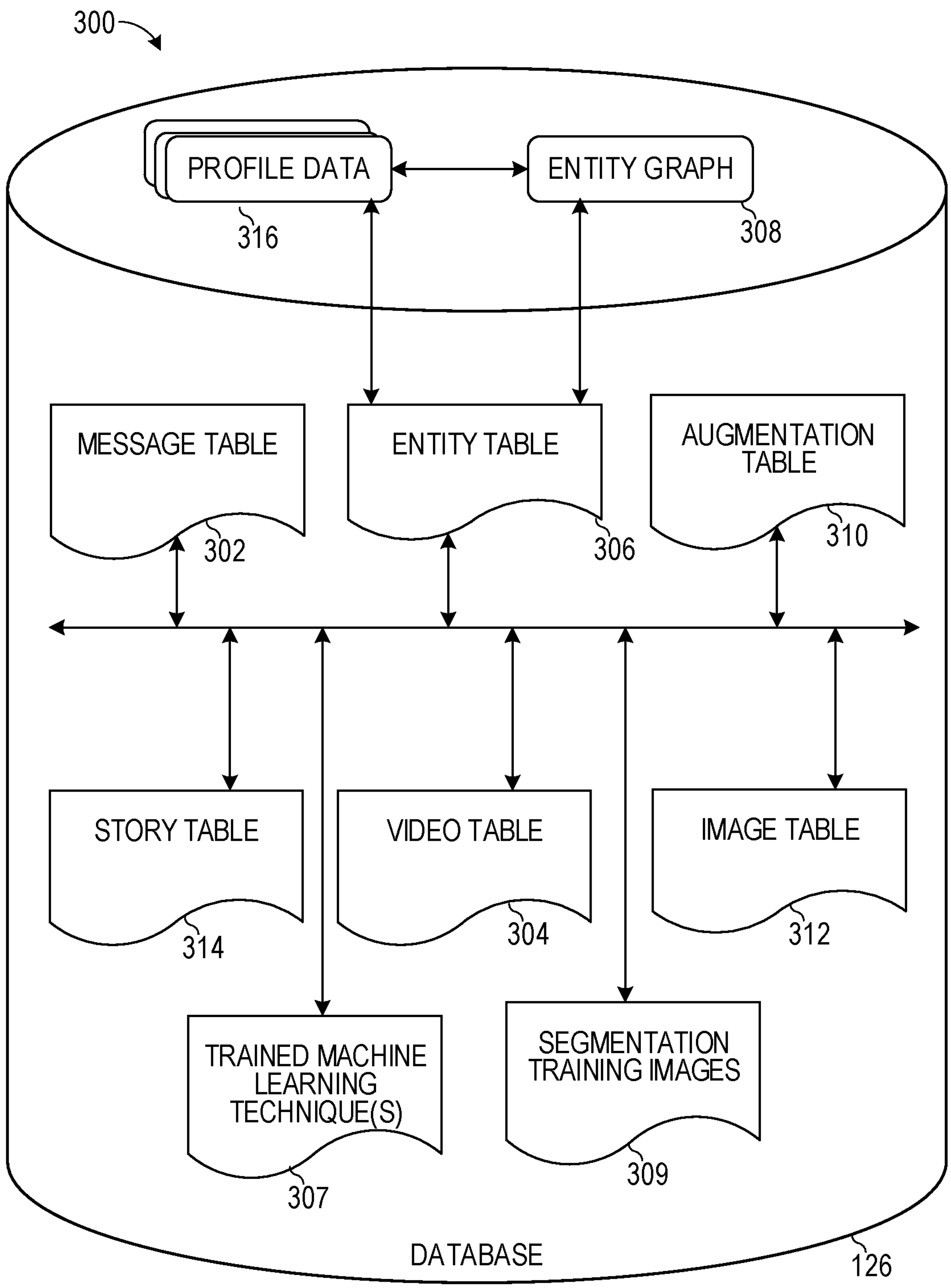
300
PROFILE DATA
316
ENTITY GRAPH
308
MESSAGE TABLE
302
ENTITY TABLE
306
AUGMENTATION TABLE
310
STORY TABLE
314
VIDEO TABLE
304
IMAGE TABLE
312
TRAINED MACHINE LEARNING TECHNIQUE(S)
307
SEGMENTATION TRAINING IMAGES
309
DATABASE
126

FIG. 3

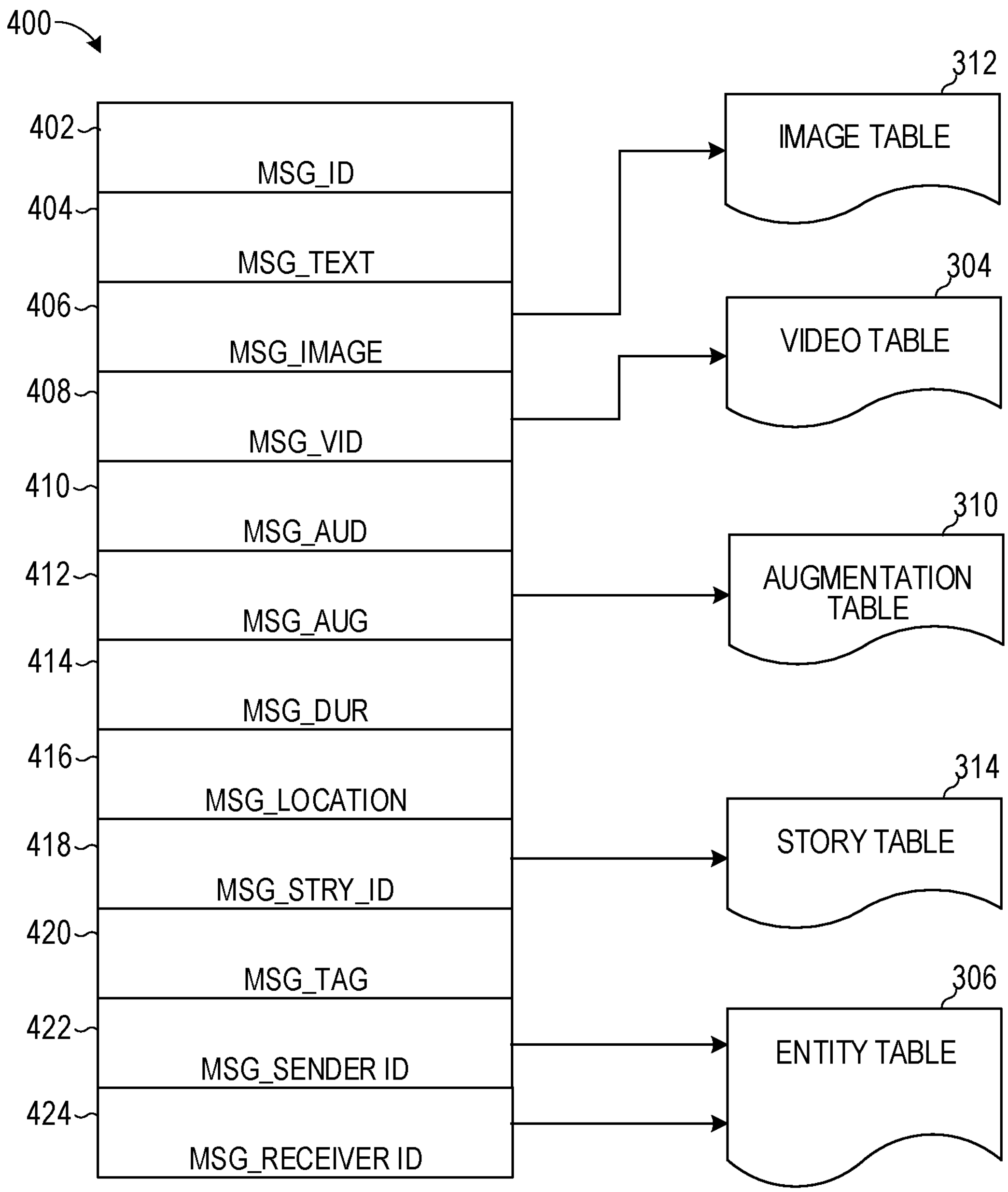
400
402
MSG_ID
404
MSG_TEXT
406
MSG_IMAGE
408
MSG_VID
410
MSG_AUD
412
MSG_AUG
414
MSG_DUR
416
MSG_LOCATION
418
MSG_STRY_ID
420
MSG_TAG
422
MSG_SENDER ID
424
MSG_RECEIVER ID
312
IMAGE TABLE
304
VIDEO TABLE
310
AUGMENTATION TABLE
314
STORY TABLE
306
ENTITY TABLE

FIG. 4

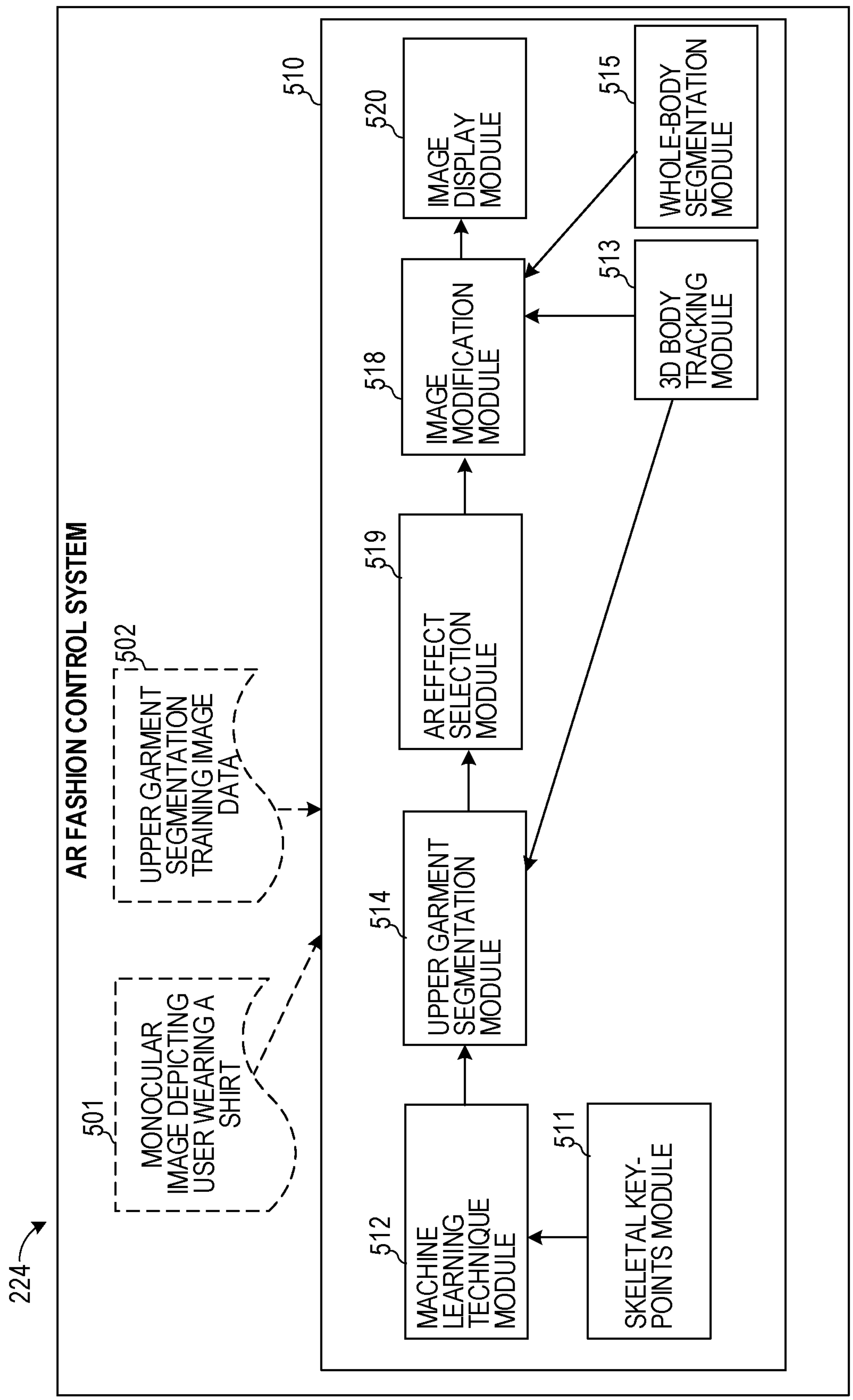
224
AR FASHION CONTROL SYSTEM
501
MONOCULAR IMAGE DEPICTING USER WEARING A SHIRT
502
UPPER GARMENT SEGMENTATION TRAINING IMAGE DATA
510
512
MACHINE LEARNING TECHNIQUE MODULE
511
SKELETAL KEY-POINTS MODULE
514
UPPER GARMENT SEGMENTATION MODULE
519
AR EFFECT SELECTION MODULE
518
IMAGE MODIFICATION MODULE
513
3D BODY TRACKING MODULE
520
IMAGE DISPLAY MODULE
515
WHOLE-BODY SEGMENTATION MODULE

FIG. 5

600

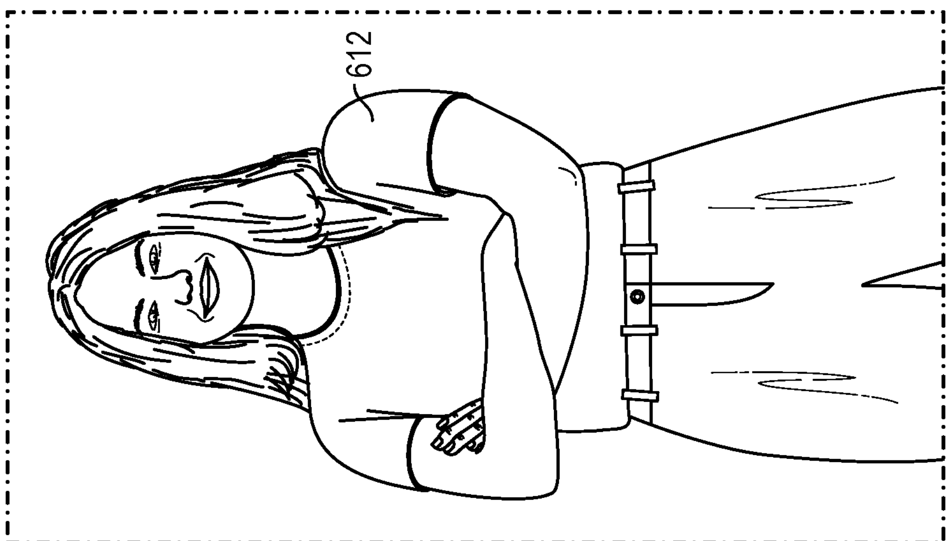

FIG. 6

700

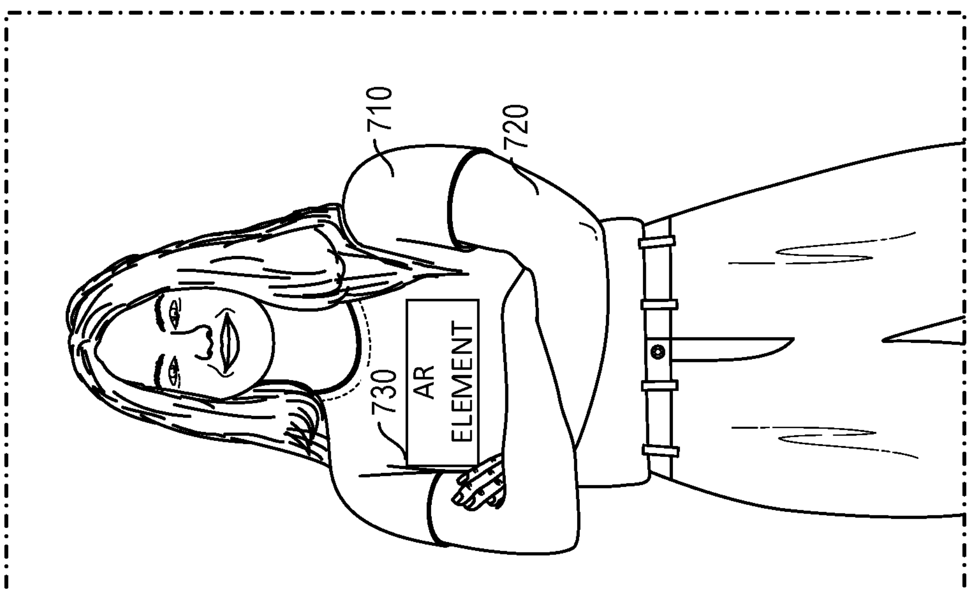

FIG. 7

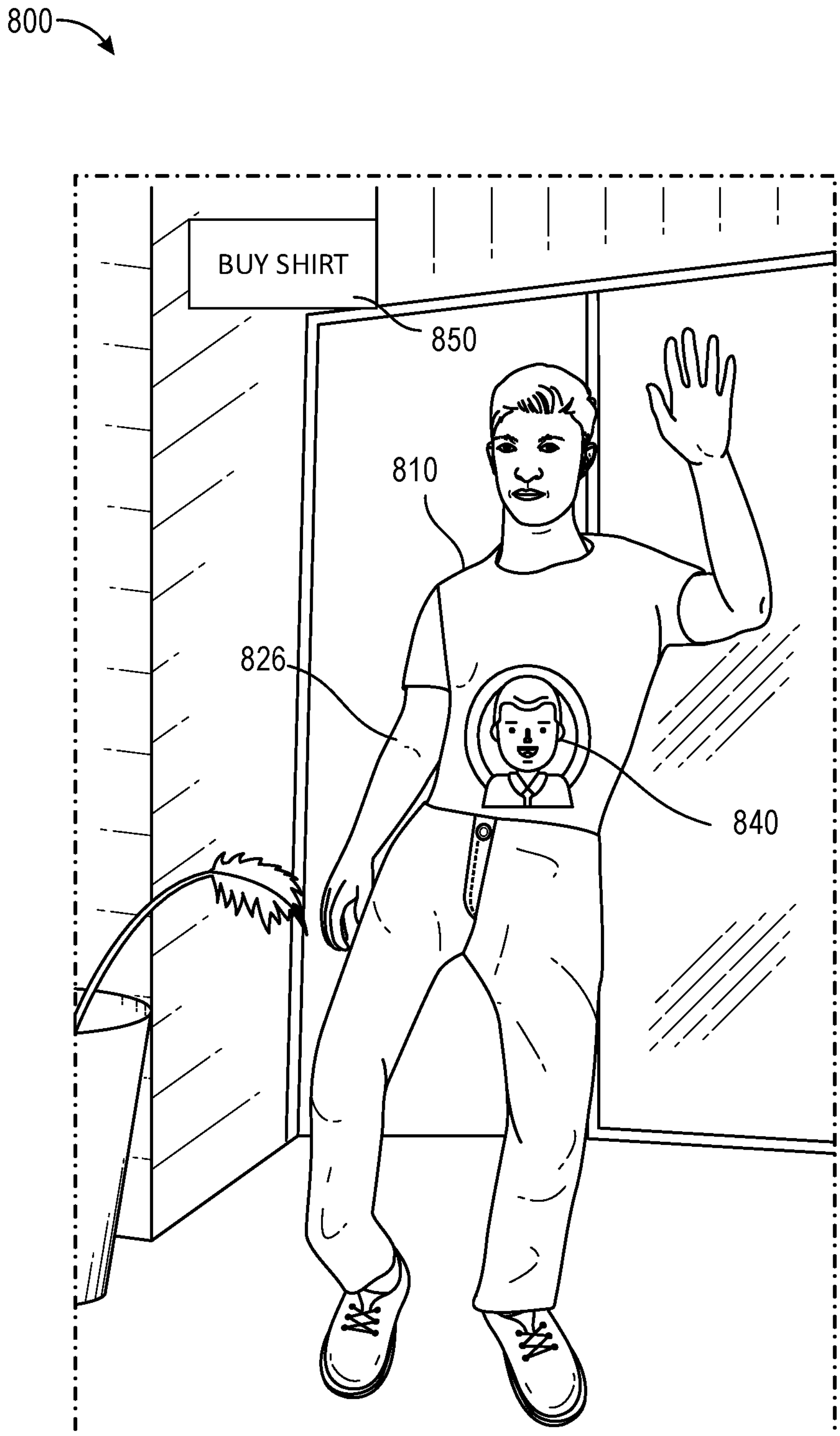

FIG. 8

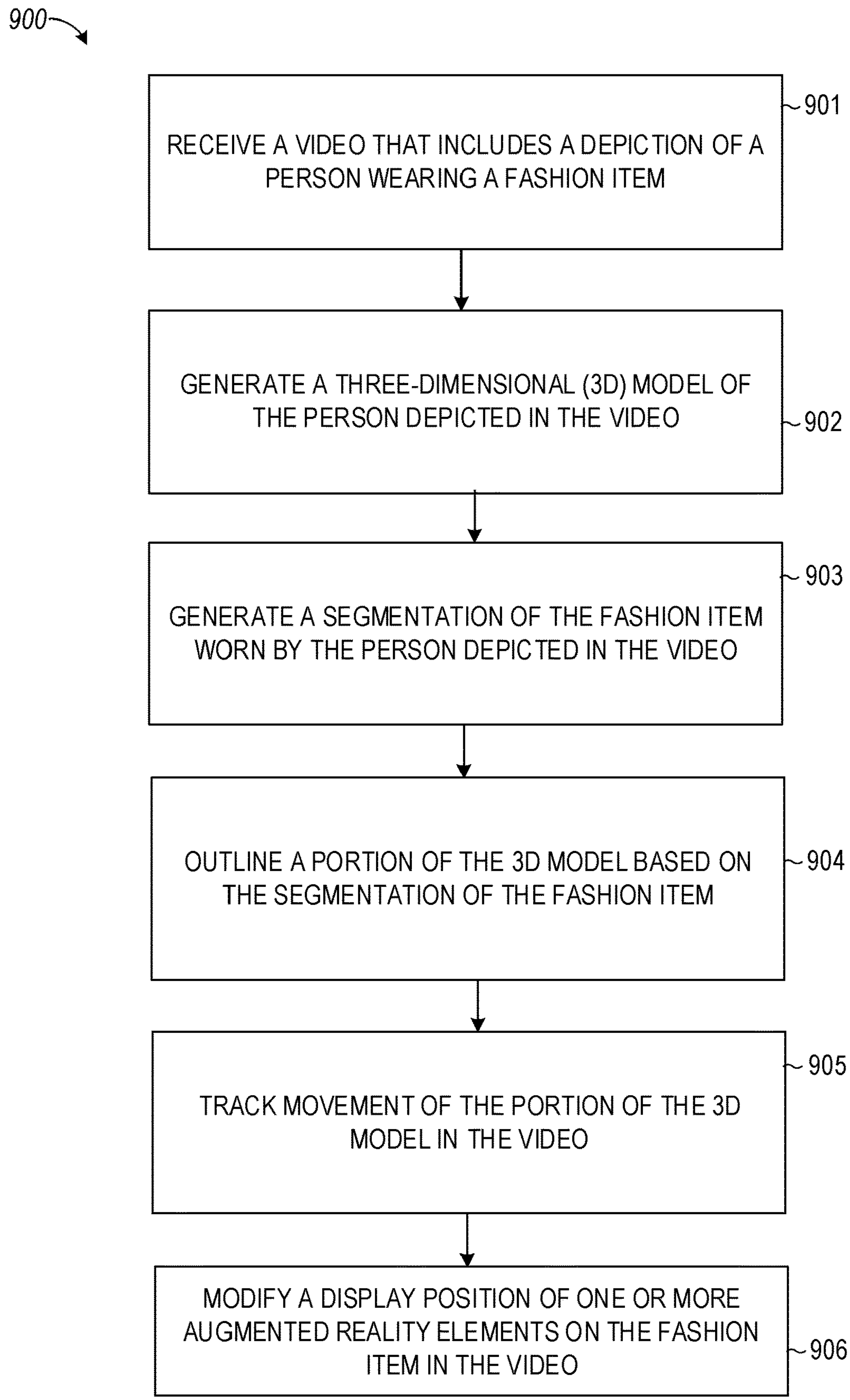
900
RECEIVE A VIDEO THAT INCLUDES A DEPICTION OF A PERSON WEARING A FASHION ITEM
901
GENERATE A THREE-DIMENSIONAL (3D) MODEL OF THE PERSON DEPICTED IN THE VIDEO
902
GENERATE A SEGMENTATION OF THE FASHION ITEM WORN BY THE PERSON DEPICTED IN THE VIDEO
903
OUTLINE A PORTION OF THE 3D MODEL BASED ON THE SEGMENTATION OF THE FASHION ITEM
904
TRACK MOVEMENT OF THE PORTION OF THE 3D MODEL IN THE VIDEO
905
MODIFY A DISPLAY POSITION OF ONE OR MORE AUGMENTED REALITY ELEMENTS ON THE FASHION ITEM IN THE VIDEO
906

FIG. 9

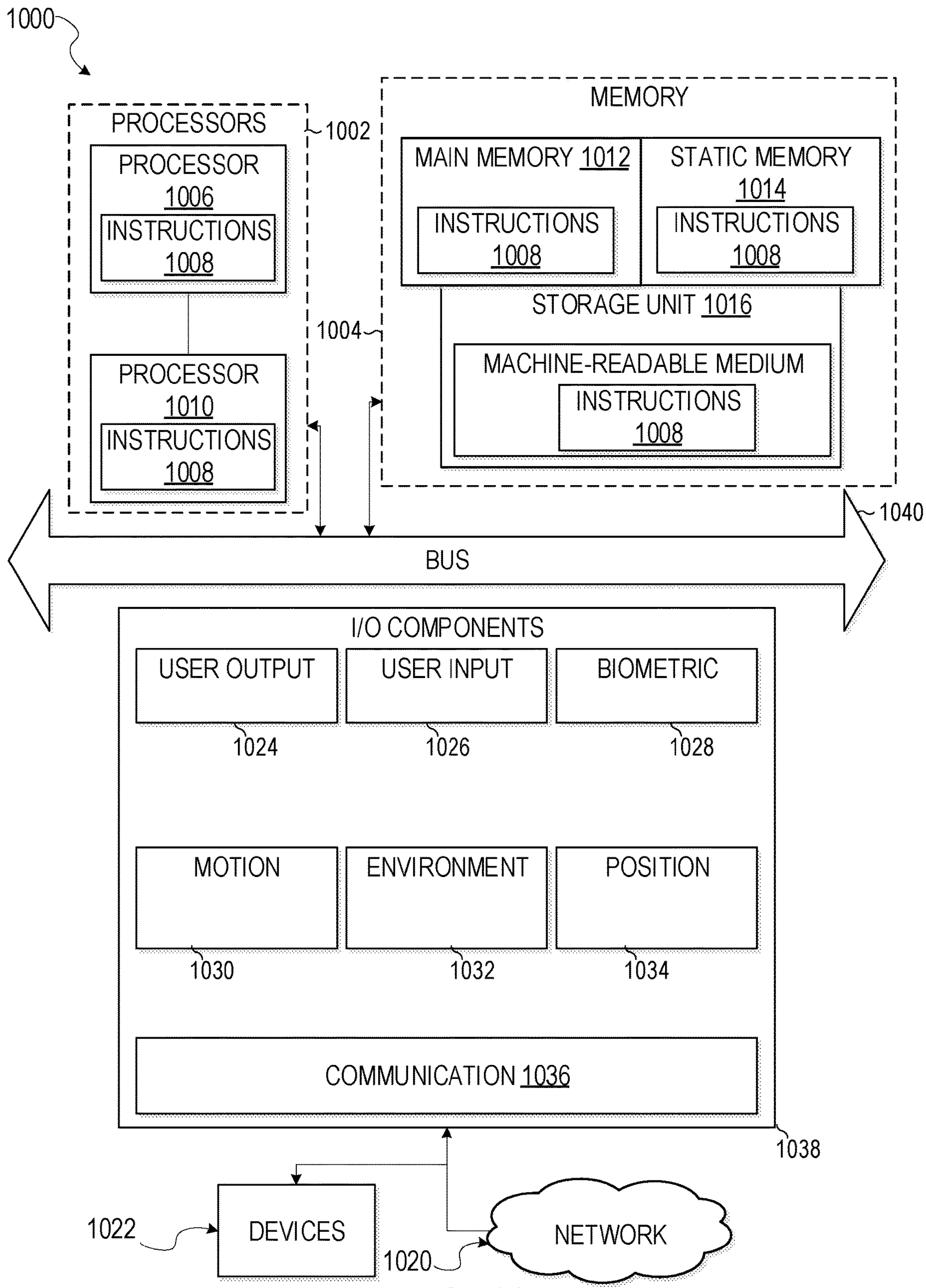
1000
PROCESSORS
1002
PROCESSOR 1006
INSTRUCTIONS 1008
PROCESSOR 1010
INSTRUCTIONS 1008
1004
MEMORY
MAIN MEMORY 1012
INSTRUCTIONS 1008
STATIC MEMORY 1014
INSTRUCTIONS 1008
STORAGE UNIT 1016
MACHINE-READABLE MEDIUM
INSTRUCTIONS 1008
1040
BUS
I/O COMPONENTS
USER OUTPUT
1024
USER INPUT
1026
BIOMETRIC
1028
MOTION
1030
ENVIRONMENT
1032
POSITION
1034
COMMUNICATION 1036
1038
1022
DEVICES
1020
NETWORK

FIG. 10

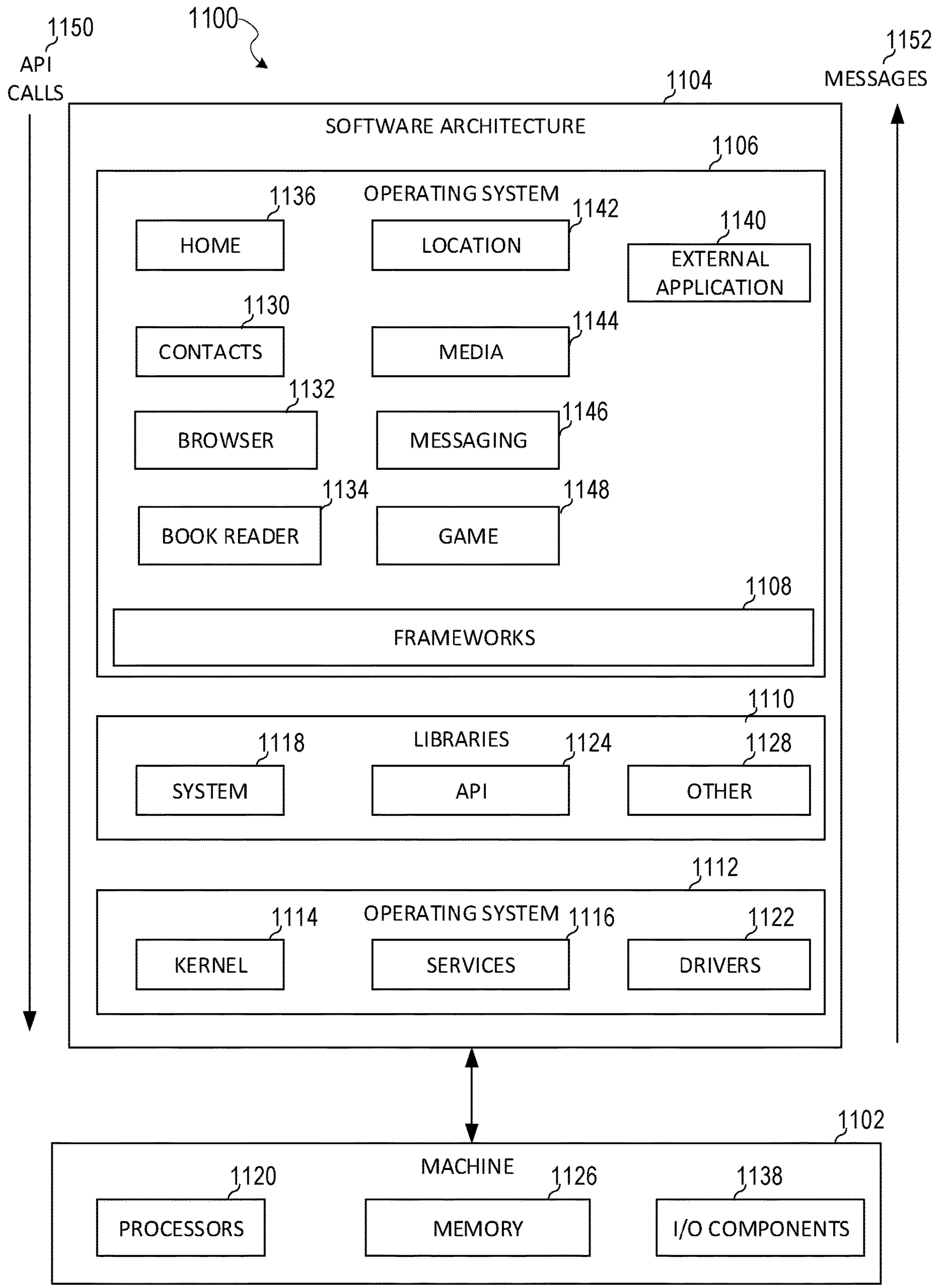
1100
1150
API CALLS
1152
MESSAGES
1104
SOFTWARE ARCHITECTURE
1106
OPERATING SYSTEM
1136
HOME
1142
LOCATION
1140
EXTERNAL APPLICATION
1130
CONTACTS
1144
MEDIA
1132
BROWSER
1146
MESSAGING
1134
BOOK READER
1148
GAME
1108
FRAMEWORKS
1110
LIBRARIES
1118
SYSTEM
1124
API
1128
OTHER
1112
OPERATING SYSTEM
1114
KERNEL
1116
SERVICES
1122
DRIVERS
1102
MACHINE
1120
PROCESSORS
1126
MEMORY
1138
I/O COMPONENTS

FIG. 11

3D UPPER GARMENT TRACKING

PRIORITY CLAIM

This application is a continuation of U.S. patent application Ser. No. 17/490,167, filed on Sep. 30, 2021, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates generally to providing augmented reality experiences using a messaging application.

BACKGROUND

Augmented reality (AR) is a modification of a virtual environment. For example, in Virtual Reality (VR), a user is completely immersed in a virtual world, whereas in AR, the user is immersed in a world where virtual objects are combined or superimposed on the real world. An AR system aims to generate and present virtual objects that interact realistically with a real-world environment and with each other. Examples of AR applications can include single or multiple player video games, instant messaging systems, and the like.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

FIG. **1** is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

FIG. **2** is a diagrammatic representation of a messaging client application, in accordance with some examples.

FIG. **3** is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

FIG. **4** is a diagrammatic representation of a message, in accordance with some examples.

FIG. **5** is a block diagram showing an example AR fashion control system, according to example examples.

FIGS. **6**, **7**, and **8** are diagrammatic representations of outputs of the AR fashion control system, in accordance with some examples.

FIG. **9** is a flowchart illustrating example operations of the AR fashion control system, according to some examples.

FIG. **10** is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

FIG. **11** is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

The description that follows includes systems, methods, techniques, instruction sequences, and computing machine

program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples. It will be evident, however, to those skilled in the art, that examples may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

Typically, VR and AR systems display images representing a given user by capturing an image of the user and, in addition, obtaining a depth map using a depth sensor of the real-world human body depicted in the image. By processing the depth map and the image together, the VR and AR systems can detect positioning of a user in the image and can appropriately modify the user or background in the images. While such systems work well, the need for a depth sensor limits the scope of their applications. This is because adding depth sensors to user devices for the purpose of modifying images increases the overall cost and complexity of the devices, making them less attractive.

Certain systems do away with the need to use depth sensors to modify images. For example, certain systems allow users to replace a background in a videoconference in which a face of the user is detected. Specifically, such systems can use specialized techniques that are optimized for recognizing a face of a user to identify the background in the images that depict the user's face. These systems can then replace only those pixels that depict the background so that the real-world background is replaced with an alternate background in the images. However, such systems are generally incapable of recognizing a whole body of a user. As such, if the user is more than a threshold distance from the camera such that more than just the face of the user is captured by the camera, the replacement of the background with an alternate background begins to fail. In such cases, the image quality is severely impacted, and portions of the face and body of the user can be inadvertently removed by the system as the system falsely identifies such portions as belonging to the background rather than the foreground of the images. Also, such systems fail to properly replace the background when more than one user is depicted in the image or video feed. Because such systems are generally incapable of distinguishing a whole body of a user in an image from a background, these systems are also unable to apply visual effects to certain portions of a user's body, such as articles of clothing.

The disclosed techniques improve the efficiency of using the electronic device by segmenting articles of clothing, fashion items, or garments worn by a user depicted in an image or video, such as a shirt worn by the user depicted in the image in addition to creating a whole-body model of the user depicted in the image or video. By segmenting the articles of clothing, fashion items, or garments worn by a user or worn by different respective users depicted in an image and creating the whole-body model, the disclosed techniques can apply one or more visual effects to the image or video, such as one or more AR elements. In an example, the disclosed techniques can apply one or more game-based AR elements to a shirt depicted in the image or video and then allow the user to interact with a game or provide a gaming experience to the user depicted in the image or video based on gestures or poses performed by the user.

In an example, the disclosed techniques apply a machine learning technique to generate a segmentation of a shirt (or upper garment) worn by a user depicted in an image (e.g., to distinguish pixels corresponding to the shirt or multiple garments worn by the user from pixels corresponding to a background of the image or a user's body parts). In this way, the disclosed techniques can apply one or more visual effects to the shirt worn by a user that has been segmented in the current image. Also, by generating the segmentation of the shirt, a position/location of the shirt in a video feed can be tracked independently or separately from positions of a user's body parts, such as a hand.

As a result, a realistic display is provided that shows the user wearing a shirt (or upper garment) while also presenting AR elements on the shirt in a way that is intuitive for the user to interact with and select. As used herein, "article of clothing," "fashion item," and "garment" are used interchangeably and should be understood to have the same meaning. This improves the overall experience of the user in using the electronic device. Also, by performing such segmentations without using a depth sensor, the overall amount of system resources needed to accomplish a task is reduced.

Networked Computing Environment

FIG. **1** is a block diagram showing an example messaging system **100** for exchanging data (e.g., messages and associated content) over a network. The messaging system **100** includes multiple instances of a client device **102**, each of which hosts a number of applications, including a messaging client **104** and other external applications **109** (e.g., third-party applications). Each messaging client **104** is communicatively coupled to other instances of the messaging client **104** (e.g., hosted on respective other client devices **102**), a messaging server system **108** and external app(s) servers **110** via a network **112** (e.g., the Internet). A messaging client **104** can also communicate with locally-hosted third-party applications, such as external apps **109**, using Application Programming Interfaces (APIs).

A messaging client **104** is able to communicate and exchange data with other messaging clients **104** and with the messaging server system **108** via the network **112**. The data exchanged between messaging clients **104**, and between a messaging client **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video, or other multimedia data).

The messaging server system **108** provides server-side functionality via the network **112** to a particular messaging client **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client **104** or by the messaging server system **108**, the location of certain functionality either within the messaging client **104** or the messaging server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108** but to later migrate this technology and functionality to the messaging client **104** where a client device **102** has sufficient processing capacity.

The messaging server system **108** supports various services and operations that are provided to the messaging client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client **104**. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client **104**.

Turning now specifically to the messaging server system **108**, an API server **116** is coupled to, and provides a programmatic interface to, application servers **114**. The application servers **114** are communicatively coupled to a database server **120**, which facilitates access to a database **126** that stores data associated with messages processed by the application servers **114**. Similarly, a web server **128** is coupled to the application servers **114** and provides web-based interfaces to the application servers **114**. To this end, the web server **128** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The API server **116** receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application servers **114**. Specifically, the API server **116** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client **104** in order to invoke functionality of the application servers **114**. The API server **116** exposes various functions supported by the application servers **114**, including account registration; login functionality; the sending of messages, via the application servers **114**, from a particular messaging client **104** to another messaging client **104**; the sending of media files (e.g., images or video) from a messaging client **104** to a messaging server **118**, and for possible access by another messaging client **104**; the settings of a collection of media data (e.g., story); the retrieval of a list of friends of a user of a client device **102**; the retrieval of such collections; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the messaging client **104**).

The application servers **114** host a number of server applications and subsystems, including, for example, a messaging server **118**, an image processing server **122**, and a social network server **124**. The messaging server **118** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available to the messaging client **104**. Other processor- and memory-intensive processing of data may also be performed server-side by the messaging server **118**, in view of the hardware requirements for such processing.

The application servers **114** also include an image processing server **122** that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server **118**.

Image processing server **122** is used to implement scan functionality of the augmentation system **208** (shown in FIG. **2**). Scan functionality includes activating and providing one or more AR experiences on a client device **102** when an image is captured by the client device **102**. Specifically, the messaging client **104** on the client device **102** can be used to activate a camera. The camera displays one or more real-time images or a video to a user along with one or more icons or identifiers of one or more AR experiences. The user can select a given one of the identifiers to launch the corresponding AR experience or perform a desired image modification (e.g., replacing a garment being worn by a user in a video or recoloring the garment worn by the user in the video or modifying the garment based on a gesture performed by the user).

The social network server **124** supports various social networking functions and services and makes these functions and services available to the messaging server **118**. To this end, the social network server **124** maintains and accesses an entity graph **308** (as shown in FIG. **3**) within the database **126**. Examples of functions and services supported by the social network server **124** include the identification of other users of the messaging system **100** with which a particular user has relationships or is "following," and also the identification of other entities and interests of a particular user.

Returning to the messaging client **104**, features and functions of an external resource (e.g., a third-party application **109** or applet) are made available to a user via an interface of the messaging client **104**. The messaging client **104** receives a user selection of an option to launch or access features of an external resource (e.g., a third-party resource), such as external apps **109**. The external resource may be a third-party application (external apps **109**) installed on the client device **102** (e.g., a "native app") or a small-scale version of the third-party application (e.g., an "applet") that is hosted on the client device **102** or remote of the client device **102** (e.g., on third-party servers **110**). The small-scale version of the third-party application includes a subset of features and functions of the third-party application (e.g., the full-scale, native version of the third-party standalone application) and is implemented using a markup-language document. In one example, the small-scale version of the third-party application (e.g., an "applet") is a web-based, markup-language version of the third-party application and is embedded in the messaging client **104**. In addition to using markup-language documents (e.g., a .*ml file), an applet may incorporate a scripting language (e.g., a .*js file or a .json file) and a style sheet (e.g., a .*ss file).

In response to receiving a user selection of the option to launch or access features of the external resource (external app **109**), the messaging client **104** determines whether the selected external resource is a web-based external resource or a locally-installed external application. In some cases, external applications **109** that are locally installed on the client device **102** can be launched independently of and separately from the messaging client **104**, such as by selecting an icon, corresponding to the external application **109**, on a home screen of the client device **102**. Small-scale versions of such external applications can be launched or accessed via the messaging client **104** and, in some examples, no or limited portions of the small-scale external application can be accessed outside of the messaging client **104**. The small-scale external application can be launched by the messaging client **104** receiving, from an external app(s) server **110**, a markup-language document associated with the small-scale external application and processing such a document.

In response to determining that the external resource is a locally-installed external application **109**, the messaging client **104** instructs the client device **102** to launch the external application **109** by executing locally-stored code corresponding to the external application **109**. In response to determining that the external resource is a web-based resource, the messaging client **104** communicates with the external app(s) servers **110** to obtain a markup-language document corresponding to the selected resource. The messaging client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the messaging client **104**.

The messaging client **104** can notify a user of the client device **102**, or other users related to such a user (e.g., "friends"), of activity taking place in one or more external resources. For example, the messaging client **104** can provide participants in a conversation (e.g., a chat session) in the messaging client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using a respective messaging client **104**, with the ability to share an item, status, state, or location in an external resource with one or more members of a group of users into a chat session. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the messaging client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

The messaging client **104** can present a list of the available external resources (e.g., third-party or external applications **109** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the external application **109** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

The messaging client **104** can present to a user one or more game-based AR experiences that can be controlled and presented on an article of clothing, such as a shirt, worn by a person (or user) depicted in the image. As an example, the messaging client **104** can detect a person in an image or video captured by the client device **102**. The messaging client **104** can segment an article of clothing (or fashion item), such as a shirt, in the image or video. While the disclosed examples are discussed in relation to a shirt worn by a person (or user of the client device **102**) depicted in an image or video, similar techniques can be applied to any other article of clothing or fashion item, such as a dress, pants, shorts, skirt, jacket, t-shirt, blouse, glasses, jewelry, a hat, ear muffs, and so forth.

In response to segmenting the shirt, the messaging client **104** can track the two-dimensional/three-dimensional (2D/3D) position of the shirt in the video separately from the position of the body of the person or user. This enables the messaging client **104** to present one or more game-based AR graphical elements on the shirt and allows the messaging client **104** to modify the game-based AR graphical elements based on tracking movement of a body part of the user in relation to the segmented shirt.

In one example, the messaging client **104** can generate a 3D body model of the person or user depicted in the image or video. The messaging client **104** can outline a portion of the 3D body model that corresponds to the fashion item worn by the person or user depicted in the image or video. In particular, the messaging client **104** can use the segmentation of the fashion item to identify and cut out a set of pixels of the 3D body model corresponding to the fashion item segmentation. The messaging client **104** can track movement of the set of pixels (the outlined portion of the 3D body model) and can update display positions of AR graphical elements based on movement of the set of pixels. Particularly, the messaging client **104** can determine the direction of movement of the portion of the 3D body model corresponding to the fashion item worn by the user or person depicted in the image or video. The messaging client **104** can update the location of the AR graphical elements based on the direction of movement to maintain a position of the AR graphical elements on the fashion item throughout frames of a video depicting the person or user.

As one example, the messaging client **104** can present an AR game board on a shirt worn by the user or person in the image. In such cases, the messaging client **104** can generate an avatar associated with the AR game board on the AR game board. The messaging client **104** can detect a body gesture performed by the person depicted in the image and can control movement of the avatar based on the body gesture. Specifically, the messaging client **104** can determine that the person raises their right hand and, in response, the messaging client **104** can move the avatar in a first direction. As another example, the messaging client **104** can determine that the person raises their left hand and, in response, the messaging client **104** can move the avatar in a second direction. The messaging client **104** can detect that the user leans towards a first side. In response, the messaging client **104** can move the avatar towards the same first side or towards an opposite side. Based on the current position of the avatar, the messaging client **104** can determine whether to increase or decrease a score associated with the AR game board. As the user leans towards the first side, at least a portion of the shirt worn by the user changes position. Based on detecting such movement of the portion of the body model corresponding to the portion of the shirt, the messaging client **104** can similarly adjust a display position of the AR game board displayed on the shirt.

The messaging client **104** can access a list of objectives or goals associated with the AR game board. The messaging client **104** can associate the different objectives and goals with different positions on the AR game board. In response to determining that the avatar has been advanced and moved closer towards a given position of one of the objectives and goals, the messaging client **104** can increase the score associated with the AR game board. The messaging client **104** can obtain a score value for each objective and goal. In response to the avatar reaching the given objective and goal, the messaging client **104** can retrieve the score value and increment the current score by the retrieved score value.

In an example, the messaging client **104** can generate an avatar for display on the shirt (or fashion item) worn by the person or user depicted in the image or video. The messaging client **104** can select a random pose for the avatar or can access a list of avatar poses associated with an AR gaming experience. The messaging client **104** can obtain a score associated with the selected pose and can adjust a pose of the avatar to match the selected pose. The messaging client **104** can present a timer counting down from a specified value (e.g., counting down from 10 seconds). The messaging client **104** can instruct the user or person to perform or mirror the pose of the avatar. The messaging client **104** can apply a body tracking process to determine whether a pose performed by the person or user matches the pose of the avatar. In response to detecting that the pose of the person or user matches the avatar pose, the messaging client **104** accesses the score associated with the selected avatar pose and increments the current score of the user or person. In response to determining that the pose of the user fails to match the pose of the avatar within the timer period (e.g., before the timer expires), the messaging client **104** does not update the score of the user or person and can decrement the score. The messaging client **104** can then access a new pose from a list of poses and can modify the pose of the avatar based on the accessed new pose. The messaging client **104** sets the timer to the specified value and begins counting down while the person or user again tries to mirror or match the new avatar pose. As the user changes poses, at least a portion of the shirt worn by the user changes position. Based on detecting such movement of the portion of the body model corresponding to the portion of the shirt, the messaging client **104** can similarly adjust a display position of the avatar displayed on the shirt.

In an example, the messaging client **104** can present an AR gaming interface on a display associated with the client device **102**. Specifically, the messaging client **104** can present a gaming application interface on an external display screen or on the same display on which an image or video of the user or person wearing the shirt is being presented. The messaging client **104** can obtain gaming controller information for the AR gaming interface. The gaming controller information can be different for different types of games. In some cases, the same controller information can be used for all types of AR games. The controller information can specify visual aspects of the gaming controller and the layout of different buttons or keys. The messaging client **104** generates an AR gaming controller based on the gaming controller information. The AR gaming controller can have the same look as a physical controller associated with the AR gaming interface. The messaging client **104** can render the AR gaming controller for display on the shirt or article of clothing worn by the user or person depicted in the image. The AR gaming controller can include a plurality of AR buttons. Based on detecting movement of the portion of the body model corresponding to the portion of the shirt, the messaging client **104** can similarly adjust a display position of the AR gaming interface displayed on the shirt.

The messaging client **104** can track movements of a body part of the user (e.g., a hand of the user) in the image or video relative to the AR gaming controller. The messaging client **104** can detect overlap between the body part and a region of the AR gaming controller that corresponds to a first AR button of a plurality of AR buttons. In response, the messaging client **104** can obtain a function or instruction associated with the first AR button, such as from the controller information associated with the AR gaming controller. The messaging client **104** can transmit an instruction to the gaming application interface that includes the function or instruction associated with the first AR button. In this way, the gaming application interface performs the function or instruction associated with the first AR button. For example, the first AR button can be a trigger associated with an AR weapon. The messaging client **104** can then instruct the gaming application interface to activate or fire a round of ammunition from the AR weapon in response to the hand of the user overlapping the region of the shirt over which the first AR button is presented.

The messaging client **104** can detect overlap between the body part and a second region of the AR gaming controller that corresponds to a second AR button of the plurality of AR buttons. In response, the messaging client **104** can obtain a second function or instruction associated with the second AR button, such as from the controller information associated with the AR gaming controller. The messaging client **104** can transmit an instruction to the gaming application interface that includes the second function or instruction associated with the second AR button. In this way, the gaming application interface performs the second function or instruction associated with the second AR button. For example, the second AR button can be a jump associated with an AR avatar or character. The messaging client **104** can then instruct the gaming application interface to cause the avatar or character to jump in response to the hand of the user overlapping the second region of the shirt over which the second AR button is presented.

In an example, the messaging client **104** can present a pong game on the fashion item. Specifically, the messaging client **104** can present an AR ball on a game board presented on a fashion item worn by the person or user depicted in the image or video. The messaging client **104** can animate movement of the AR ball (steer the ball) around the game board based on 3D movement of the body or body part (e.g., hand or hands) of the person or user depicted in the image or video. For example, the messaging client **104** can animate the AR ball moving or bouncing from a first side of the fashion item to a second side of the fashion item based on movement or positioning of the body part(s) of the person or user. Specifically, the AR ball can be animated as moving along a first trajectory towards a first side of the shirt worn by the user or person depicted in the image or video. The messaging client **104** can detect a position of a left hand of the user along an edge of the first side of the shirt. The messaging client **104** can determine that the AR ball contacts the left hand on the edge of the first side of the shirt. In response, the messaging client **104** can bounce the AR ball along a second trajectory towards a second side opposite the first side of the shirt and also increment a score associated with the user or person. If the left hand of the user does not contact or overlap the position of the AR ball on the first side of the shirt, the messaging client **104** can animate the AR ball as falling off the shirt and can reduce a score associated with the user or person depicted in the image or video. The messaging client **104** can detect a position of a right hand of the user along an edge of the second side of the shirt. The messaging client **104** can determine that the AR ball contacts the right hand on the edge of the second side of the shirt. In response, the messaging client **104** can bounce the AR ball along a third trajectory towards the first side of the shirt and also increment a score associated with the user or person. If the right hand of the user does not contact or overlap the position of the AR ball on the second side of the shirt, the messaging client **104** can animate the AR ball as falling off the shirt and can reduce a score associated with the user or person depicted in the image or video.

Based on detecting movement of the portion of the body model corresponding to the portion of the shirt, the messaging client **104** can similarly adjust a display position of the AR ball that is moved on the shirt. Namely, the AR ball can be moved based on both an animation trajectory of the ball and physical movement of the person depicted in the image or video so as to maintain the movement of the AR ball along the same trajectory and portions of the shirt as the person moves around. For example, the AR ball can be animated as moving along a first direction from a first side of the shirt to a second side of the shirt. While moving along the first direction, the messaging client **104** can detect that the 3D body model has shifted in position to lean towards the second side by 10 degrees or some other amount relative to an initial position when the AR ball was moving along the first direction. In response, the messaging client **104** can change the first direction by 10 degrees to update the movement of the AR ball in accordance with the change in the 3D body model position. This results in maintaining the movement of the AR ball from the first side of the shirt to the second side of the shirt along the first direction.

In some cases, the messaging client **104** can select the third trajectory or can compute the third trajectory based on where the ball contacts the right hand. If the ball contacts a top portion of the hand, the messaging client **104** can reflect the ball along a third trajectory that forms a 65 degree angle with the second trajectory. If the ball contacts a lower portion of the hand (e.g., bottom half of the hand), the messaging client **104** can reflect the ball along a third trajectory that forms a 35 degree angle with the second trajectory.

In an example, the messaging client **104** can present an AR game on the fashion item in which one or more AR objects appear to shoot out of the fashion item worn by the user or person depicted in the image or video. For example, the messaging client **104** can generate one or more AR objects (e.g., flying saucers) and can animate each one of the AR objects as coming out of a portion of the shirt along a surface normal of the shirt worn by the user or person. Each AR object can be generated and animated in succession and flying out in different directions along the surface normal of the shirt at a different rate. Each AR object can be of a different size and shape (e.g., increasing or decreasing in size and shape). The rate can increase or decrease over time and the AR objects can be set to disappear or be removed from display after a threshold period of time. The messaging client **104** can detect contact between each AR object that is animated as coming out of the shirt and a body part of the user (e.g., a hand). In response to determining that the body part contacts the AR object before the AR object disappears, the messaging client **104** increments a score associated with the AR game. The score can be incremented by a one or by some factor that is based on the type, style, size, rate, or other parameter of the AR object which has been contacted.

In an example, the messaging client **104** can present a catch game on the fashion item worn by the user depicted in the image or video. In this case, the messaging client **104** can present an AR net or hole on a portion of a shirt or fashion item. The size of the net or hole can be specified as a factor of the level of difficulty of the catch game (smaller nets/holes result in greater level of difficulty than larger nets/holes). The messaging client **104** can generate one or more AR objects (e.g., flying saucers) and can animate each one of the AR objects as flying towards the user or person depicted in the image or video along a surface normal of the shirt worn by the user or person. Each AR object can be generated and animated in succession and flying towards the user or person in different directions along the surface normal of the shirt at a different rate. Each AR object can be of a different size and shape (e.g., increasing or decreasing in size and shape). The rate can increase or decrease over time and the AR objects can be set to disappear or removed from display after a threshold period of time. The messaging client **104** can detect contact between each AR object that is animated as heading or flying towards the user or person and the AR net or hole. For example, the user or person can move around in 3D in the image or video to attempt to line up the AR net or hole with the flying AR object. In response to determining that the AR net or hole contacts the AR object before the AR object disappears, the messaging client **104** increments a score associated with the AR game. The score can be incremented by a one or by some factor that is based on the type, style, size, rate, or other parameter of the AR object which has been contacted.

In an example, the messaging client **104** presents an AR drawing board on the fashion item. The messaging client **104** can detect movement of a body part of the person depicted in the image or video. In response to detecting that the body part overlaps the AR drawing board, the messaging client **104** adds one or more AR graphics to the AR drawing board. For example, the messaging client **104** can draw or paint a line on the AR drawing board along a direction of the body part that is detected to move over the position of the AR drawing board. The messaging client **104** can start and stop the drawing of the line based on a pose of the body part that overlaps the portion of the fashion item. For example, if the user's hand is in a closed first pose, the messaging client **104** can draw a line that tracks the position of the hand along the AR drawing board. If the user's hand transitions to an open first pose, the messaging client **104** stops drawing the line even though the hand still overlaps the AR drawing board. This allows the user to seamlessly and easily control drawing and writing on the AR drawing board on the fashion item.

In an example, the messaging client **104** can detect movement of the body part of the user and can adjust a shape of one or more AR gaming elements or line based on the body part. For example, the messaging client **104** can determine that a right hand is holding two fingers up. In response, the messaging client **104** can increase a weight of the line being used to draw on the AR drawing board. As more fingers are raised, the line weight is increased or decreased. Similarly, a color of the line can be changed based on a hand configuration (e.g., the number of fingers being raised).

In an example, the messaging client **104** can detect one or more wrinkles on the fashion item worn by the person depicted in the image. The messaging client **104** can modify movement of one or more game-based AR elements based on the position of the game-based AR elements relative to a location of the wrinkles. For example, an AR ball can move from one side of the shirt to another and can be navigated around AR mountains that have heights and locations based on the wrinkles. If the AR ball contacts a given AR mountain, the speed and trajectory of the ball is changed based on the height of the mountain. In another example, an AR car can be moved based on body movement of the user depicted in the image or video. A speed of the car is decreased as the car starts climbing up a given one of the AR mountains and the speed of the car (rate at which the car is moved along the shirt) increases as the car moves past the peak of the mountain and heads down the mountain.

In one example, the ball, car, avatar, remote control, or any other AR gaming element discussed in this disclosure can be generated as a unique asset for the user based on a non-fungible token associated with the asset. For example, the messaging client **104** can allow the user to purchase a given gaming element and can associate a non-fungible token with the gaming element in response to the user completing a purchase transaction for the gaming element. The messaging client **104** can add the purchased gaming element to the user's account and can add the gaming element as a unique asset to the AR game presented on the article of clothing. In an example, the messaging client **104** allows the user to create unique fashion items based on non-fungible tokens. For example, the messaging client **104** can capture a screenshot or a video that includes the fashion item worn by the user depicted in the image or video and that includes or is combined with the one or more game-based AR elements. The messaging client **104** can generate an AR element that represents or includes the fashion item worn by the user depicted in the image or video combined with the one or more game-based AR elements. The messaging client **104** can generate a non-fungible token and associate the non-fungible token with the AR element. This allows the user or person to exchange or sell the AR element as a unique asset with one or more other users.

As one example, the messaging client **104** can generate text for display on the shirt worn by the person depicted in the image or video. The text can be rendered on the shirt worn by the person in response to detecting voice input from a person depicted in the image or video or other user; or words of existing text can be replaced with other words based on the voice input of the person depicted in the image or video or other user. For example, the messaging client **104** can receive voice input associated with the person depicted in the image or video, such as via a microphone of the client device **102**. In response to receiving the voice input, the messaging client **104** can apply one or more machine learning techniques to features or characteristics of the voice input to detect a particular emotion or mood, such as happy, sad, surprised, confused, upset, and so forth associated with the voice input. In some examples, the messaging client **104** can search a database of words to identify one or more words that are associated with the emotion or mood of the voice input. The messaging client **104** can select a subset of the identified words, such as based on a rank, popularity, user profile, randomness, uniqueness, or any combination thereof. The messaging client **104** can then add an AR element that includes text to the shirt worn by the person in the image or video. The messaging client **104** can include, in the text of the AR element, the selected subset of the identified words.

In some examples, the messaging client **104** can search a database of graphics (e.g., emojis or avatars) to identify one or more graphics that are associated with the emotion or mood of the voice input. The messaging client **104** can select a subset of the identified graphics, such as based on a rank, popularity, user profile, randomness, uniqueness, or any combination thereof. The messaging client **104** can then add an AR element that includes graphics to the shirt worn by the person in the image or video. The messaging client **104** can include, in the graphics of the AR element, the selected subset of the identified graphics.

In another implementation, the messaging client **104** can apply one or more known techniques to transcribe the voice input to generate a transcription of the voice input that includes one or more words. The messaging client **104** can generate an AR element that includes the one or more words of the transcription and add the AR element to the shirt worn by the person in the image or video.

In some implementations, the messaging client **104** can detect a phrase written physically on the shirt worn by the person depicted in the image or video. The messaging client **104** can perform word recognition to the phrase to identify a word associated with an adjective that describes an emotion or mood. In response to identifying the word associated with the adjective, the messaging client **104** can generate an AR version of the identified word and include text in the AR version that represents the voice input. The messaging client **104** can replace the physical word with the AR version of the word, such as by overlaying the AR version of the word on top of the physical word that is on the shirt worn by the person depicted in the image or video. In this way, the messaging client **104** can generate new text or modify existing text on the shirt worn by the person depicted in the image or video to represent a voice input emotion or characteristic of the person depicted in the image or video. As the characteristics or features of the voice input changes in real time (e.g., from being associated with a happy emotion or mood to being associated with an angry emotion or mood), the messaging client **104** can update the AR words displayed on the shirt worn by the person depicted in the image or video to represent different characteristics or features of the voice input.

In some implementations, the messaging client **104** can detect an emblem or logo drawn physically on the shirt worn by the person depicted in the image or video. The messaging client **104** can detect a command in the voice input received in association with the person depicted in the image or video. The messaging client **104** can generate an AR version of the emblem or logo based on the command and can replace the physical emblem or logo with the AR version of the emblem or logo, such as by overlaying the AR version of the emblem or logo on top of the physical emblem or logo that is on the shirt worn by the person depicted in the image or video. In this way, the messaging client **104** can modify an existing emblem or logo on the shirt worn by the person depicted in the image or video to represent a command in the voice input associated with the person depicted in the image or video.

As another example, the messaging client **104** can adjust an expression of an AR avatar or emoji depicted on a shirt worn by a person in the image or video to match the characteristics or features of the voice input (e.g., the emotion or mood of the voice input) associated with the person depicted in the image or video. For example, the messaging client **104** can detect voice input associated with the person depicted in the image or video. In response to detecting the voice input, the messaging client **104** can apply one or more machine learning techniques to features or characteristics of the voice input to detect a particular emotion or mood, such as happy, sad, surprised, confused, upset, and so forth associated with the voice input. The messaging client **104** can search a database of avatar or emoji expressions that are associated with the emotion or mood of the voice input. The messaging client **104** can then adjust a facial expression of an avatar or emoji displayed on the shirt worn by the person depicted in the image or video to match or correspond to the characteristics of the voice input associated with the person depicted in the image or video.

As one example, the messaging client **104** can adjust a fashion item pattern, type, or style based on voice input received in association with a person depicted in an image or video. For example, the messaging client **104** can process the voice input associated with the person depicted in an image or video. The messaging client **104** can determine that the voice input includes a command or request to change a style, pattern, or type of the fashion item worn by the person depicted in the image or video. For example, the voice input can request to change a shirt worn by the person depicted in the image or video to a jacket. As another example, the voice input can request to change a skirt worn by the person depicted in the image or video to a dress. As another example, the voice input can request to change shorts worn by the person depicted in the image or video to pants. The fashion item or clothing can be physically applied to the body of the person depicted in the image or can be applied in AR using an AR try-on experience. The messaging client **104** can then select a given one of the fashion item patterns, styles, or types associated with the voice input (e.g., an AR jacket when the voice input requests to change a shirt to a jacket) to generate an AR graphical element associated with the selected fashion item pattern, style, or type. The messaging client **104** applies the AR graphical element to the fashion item worn by the person depicted in the image or video. As a result, the messaging client **104** outputs or generates a display in which the person is depicted in the image or video wearing the AR graphical element (e.g., the AR jacket) instead of the real-world fashion item (e.g., the shirt).

As one example, the messaging client **104** can adjust a fashion item color or pattern based on voice input received in association with a person depicted in an image or video. For example, the messaging client **104** can process the voice input associated with the person depicted in an image or video. The messaging client **104** can determine that the voice input includes a command or request to change a color or pattern (from a solid color to polka dots) of the fashion item (e.g., the shirt) worn by the person depicted in the image or video. The messaging client **104** can then determine a color or pattern associated with the voice input. The messaging client **104** can populate a graphic having a shape and size of the segmentation of the fashion item (e.g., the shirt) and including the color or pattern specified by the voice input to generate an AR graphical element. Namely, the messaging client **104** can obtain the segmentation of the fashion item worn by the person depicted in the image or video and generate a graphic that has the same size and shape as the segmentation and which is filled with the color or pattern specified by the voice input. The messaging client **104** applies the AR graphical element to the fashion item worn by the person depicted in the image or video. As a result, the messaging client **104** outputs or generates a display in which the color or pattern of the fashion item worn by the person depicted in the image or video changes responsive to the voice input.

As one example, the messaging client **104** can select a region of a fashion item over which to apply an AR element based on voice input received in association with a person depicted in an image or video. For example, the messaging client **104** can process the voice input associated with the person depicted in an image or video. The messaging client **104** can determine a pitch or frequency associated with the voice input. The messaging client **104** can then select a region of the fashion item based on the pitch or frequency associated with the voice input. As an example, in response to determining that the pitch or frequency is above a first threshold, the messaging client **104** can select an upper portion of the fashion item. In response to determining that the pitch or frequency is between a second threshold and the first threshold, the messaging client **104** can select a middle portion of the fashion item. In response to determining that the pitch or frequency is below the second threshold, the messaging client **104** can select a lower portion of the fashion item. The messaging client **104** can apply an AR graphical element (e.g., text or graphics) to the fashion item worn by the person depicted in the image or video at the selected portion. As the pitch or frequency of the voice input changes over time and in real time, the placement of the AR graphical element can also change in real time.

As one example, the messaging client **104** can select a region of a fashion item over which to apply an AR element based on an acoustic direction of the voice input received in association with a person depicted in an image or video. For example, the messaging client **104** can process the voice input associated with the person depicted in an image or video. The messaging client **104** can determine an acoustic direction associated with the voice input. The messaging client **104** can then select a region of the fashion item that is on a path of the acoustic direction associated with the voice input. As an example, as the user changes the positioning of their head when they speak, the acoustic direction changes and is focused towards different regions of the fashion item. The messaging client **104** can apply an AR graphical element (e.g., text or graphics) to the fashion item worn by the person depicted in the image or video at the selected region. As the acoustic direction of the voice input changes over time and in real time, the placement of the AR graphical element can also change in real time.

As one example, the messaging client **104** can display AR music-related elements. In such cases, the messaging client **104** can determine that a music file is being played on the client device **102** during capture of the monocular image or video, such as by detecting musical audio input from a device microphone. In response, the messaging client **104** can obtain lyrics associated with the music file in a synchronized manner with the music being played and generate, for display on the shirt, AR lyrics (AR music related elements) corresponding to a portion of the lyrics. In one implementation, the lyrics can be obtained over the Internet by accessing a lyrics file associated with the music file from a remote server. In another implementation, the AR lyrics can be generated in real-time by transcribing a vocals portion of the music file into the corresponding AR lyrics.

The messaging client **104** can present a first set of AR lyrics corresponding to a current playback position (e.g., playback position 1:30 (minutes:seconds)) of the music being played back to represent the currently sung portion of a song. At a later time (e.g., after a specified time interval, such as 2 seconds), the messaging client **104** can swap out the first set of AR lyrics with a second set of lyrics corresponding to a current, earlier, or later playback position (e.g., playback position 1:32 (minutes:seconds)) of the music being played back to represent the currently sung portion of a song. The AR lyrics can be presented on the center of the shirt worn by the user depicted in the image or video or at any other suitable location (e.g., the sleeves or collar). In another example, the first portion of the AR lyrics can initially be presented on a first portion of the shirt (e.g., the collar), then the first portion of the AR lyrics can remain displayed when the second portion of the AR lyrics are animated as entering a second portion of the shirt (e.g., the sleeves), then the first and second portions of the AR lyrics can remain displayed when a third portion of the AR lyrics (corresponding to a subsequent play position that follows the display positions corresponding to the first and second portions of the lyrics) are animated as entering a third portion of the shirt (e.g., the center). After the third portion of the AR lyrics are displayed on the third portion of the shirt, a fourth portion of the AR lyrics can replace the first portion of the AR lyrics, such that at that time, the fourth portion of the AR lyrics is presented together with the second and third portions of the AR lyrics.

As another example, the messaging client **104** can determine that a music file is being played on the client device **102** during capture of the monocular image or video. In response, the messaging client **104** can obtain a music score that includes music notes associated with the music file in a synchronized manner with the music being played and generate for display, on the shirt, a portion AR music notes (AR music related elements) based on the music notes or a portion of the music notes of the music score. In one implementation, the music score can be obtained over the Internet by accessing a music score associated with the music file from a remote server. In another implementation, the AR music notes of the music score can be generated in real-time by transcribing a music portion of the music file into the corresponding AR music score.

The messaging client **104** can present a first set of AR music notes (AR music related elements) corresponding to a current playback position (e.g., playback position 1:30 (minutes:seconds)) of the music being played back to represent the currently played portion of a song. At a later time (e.g., after a specified time interval, such as 2 seconds), the messaging client **104** can swap out the first set of AR music notes with a second set of music notes corresponding to a current, earlier, or later playback position (e.g., playback position 1:32 (minutes:seconds)) of the music being played back to represent the currently played portion of a song. The AR music notes can be presented on the center of the shirt worn by the user depicted in the image or video or at any other suitable location (e.g., the sleeves or collar). In another example, the first portion of the AR music notes can initially be presented on a first portion of the shirt (e.g., the collar), then the first portion of the AR music notes can remain displayed when the second portion of the music notes are animated as entering a second portion of the shirt (e.g., the sleeves), then the first and second portions of the AR music notes can remain displayed when a third portion of the AR music notes (corresponding to a subsequent play position that follows the display positions corresponding to the first and second portions of the music notes) are animated as entering a third portion of the shirt (e.g., the center). After the third portion of the AR music notes are displayed on the third portion of the shirt, a fourth portion of the music notes can replace the first portion of the AR music notes, such that at that time, the fourth portion of the music notes is presented together with the second and third portions of the AR music notes.

In one example, the messaging client **104** can measure the tempo of the music file being played on the client device **102**. The messaging client **104** can adjust the animation rhythm or rate at which the AR music notes are added/removed from the shirt based on the tempo. Specifically, if the tempo is a first value, the messaging client **104** can add/remove AR music notes at a first rate and if the tempo is a greater second value, the messaging client **104** can add/remove the AR music notes at a faster second rate.

In another example, the messaging client **104** can determine a plurality of instruments associated with the music file being played on the client device. A first of the plurality of instruments can be associated with a first portion of the music notes and a second of the plurality of instruments can be associated with a second portion of the music notes. For example, a first of the plurality of instruments can include a guitar and can correspond to sounds produced based on the first portion of the music notes. A first of the plurality of instruments can include drums (including a cymbal or accent cymbal) and can correspond to sounds produced based on the second portion of the music notes together with the first portion of the music notes or after the first portion of the music notes.

In response to determining that the first of the plurality of instruments is associated with a first type of instrument (e.g., a guitar or instrument that produces sounds in the frequency range of treble), the messaging client **104** can display the first portion of the music notes on a first fashion item (e.g., garment) that is depicted in the image or video, such as the shirt worn by the user. Similarly, in response to determining that the second of the plurality of instruments is associated with a second type of instrument (e.g., drums or instrument that produces sounds in the frequency range of bass), the messaging client **104** can display the second portion of the music notes on a second fashion item worn by the user, such as pants or glasses. In another implementation, the position of the music notes in relation to the user depicted in the image can be based on the frequency of sound produced by the music notes. Namely, a frequency of sounds in a first range that is higher than a second range can be positioned over a fashion item worn at a first location on the user's body that is higher than another fashion item worn at a second location on the user's body. In some cases, the messaging client **104** can detect a cymbal sound in a portion of the music file being played on the client device **102**. In response, the messaging client **104** can select an AR graphical element or effect (e.g., stars, a glowing effect, sparkles effect, or any other suitable effect) and generate for display that AR graphical element or effect concurrently with the cymbal sound. In one example, the AR graphical element or effect displayed concurrently with the cymbal sound can be temporarily displayed on lenses of glasses worn by the user in the image or video for a specified period of time (e.g., 3 seconds), after which the AR graphical element or effect displayed concurrently with the cymbal sound is removed from the display. This AR graphical element or effect can be presented while the music notes are presented on other fashion items worn by the user depicted in the image or video.

In another example, in response to determining that the first of the plurality of instruments is associated with a first type of instrument (e.g., a guitar or instrument that produces sounds in the frequency range of treble), the messaging client **104** can display a first type of AR graphical element (or modify a given AR element displayed on the first portion of the first fashion item according to a first rhythm or rate) on the first portion of a first fashion item (e.g., garment) that is depicted in the image or video, such as the shirt worn by the user. Similarly, in response to determining that the second of the plurality of instruments is associated with a second type of instrument (e.g., drums or instrument that produces sounds in the frequency range of bass), the messaging client **104** can display a second type of AR graphical element (or modify the given AR element displayed on the second portion of the first fashion item according to a second rhythm or rate) on the second portion of the first fashion item worn by the user. For example, a right shoulder portion of a shirt depicted in the image can change colors according to a rhythm of a guitar instrument while a left shoulder portion of the shirt can change colors according to a rhythm of a piano instrument in the music file.

In another example, the messaging client **104** can obtain an image or video of an artist associated with the music file being played on the client device **102**. The messaging client **104** can generate an AR element (AR music related elements) that includes the image or video of the artist and display that AR element on a portion of the shirt worn by the user. The messaging client **104** can detect a gesture being made by the left and right hands of the user depicted in the image or video. Based on the gesture, the messaging client **104** can adjust an aspect ratio of the AR element that includes the image or video of the artist on the shirt.

In another example, the messaging client **104** accesses a music player that includes a plurality of playback control options. The playback control options can include a music selection option, an artist selection option, a play option, a pause option, a skip option, a volume control option, music editing option (e.g., speed, pitch, or frequency change), and/or any other suitable option. The messaging client **104** can generate one or more AR music elements (AR music related elements) each corresponding to a different one of the plurality of playback control options. The messaging client **104** can apply the one or more AR music elements to the shirt worn by the user depicted in the image or video, such as by presenting the AR music elements on different portions of the shirt worn by the user. The messaging client **104** can access a video feed depicting a hand of the user and that depicts the one or more AR music elements on the shirt worn by the user. The messaging client **104** determines that a position of the hand of the user in the video feed overlaps a given one of the pluralities of playback control options that are on the shirt worn by the user, such as the pause playback option. In response, the messaging client **104** activates a music player function associated with the given one of the pluralities of playback control options, such as by pausing playback. In one implementation, a body tracking model is applied to the hand of the user depicted in the video to determine movement of the hand of the user in the video feed towards the position that overlaps the given one of the pluralities of playback control options.

In certain implementations, the messaging client **104** enables multiple directions of musical interactions with a fashion item worn by a user depicted in an image or video. For example, the messaging client **104** can enable a first interaction in which an AR element is presented on a fashion item worn by the user or person depicted in the image or video using sound as a trigger. In this case, the messaging client **104** presents AR elements on a fashion item based on characteristics of a music or sound file being accessed or played by the client device **102**. In another example, the messaging client **104** can enable a second interaction in which audio is generated based on user interaction with the fashion item worn by the user or person depicted in the image or video. Specifically, the messaging client **104** can generate different audio outputs or sounds on the basis of where a user or person depicted in the image or video places a hand or body part in relation to the fashion item worn by the user or person.

In another example, the messaging client **104** accesses a music creation application that includes a plurality of music creation options. The music creation options can include a music style control option, a music tempo control option, a musical instrument control option, a music sound generation option, and any other suitable option. The messaging client **104** can generate one or more AR music elements each corresponding to a different one of the plurality of music creation options. The messaging client **104** can apply the one or more AR music elements to the shirt worn by the user depicted in the image or video, such as by presenting the AR music AR elements on different portions of the shirt worn by the user. The messaging client **104** can access a video feed depicting a hand of the user and that depicts the one or more AR music elements on the shirt worn by the user. The messaging client **104** determines that a position of the hand of the user in the video feed overlaps a given one of the pluralities of music creation options that are on the shirt worn by the user, such as the music sound generation option. In response, the messaging client **104** generates an audio signal or modifies the current music file being played on the client device **102** based on a music note associated with the given one of the pluralities of music creation options. In one implementation, a body tracking model is applied to the hand of the user depicted in the video to determine movement of the hand of the user in the video feed towards the position that overlaps the given one of the pluralities of playback control options.

In one example, the hand of the user can be detected as overlapping a left sleeve portion of the shirt over which a first AR music element is displayed. In response, a first music creation option is selected and activated to generate an audio signal corresponding to the first music creation option. In another example, the hand of the user can be detected as overlapping a right sleeve portion of the shirt over which a second AR music element is displayed. In response, a second music creation option is selected and activated to generate an audio signal corresponding to the second music creation option. As a result, the messaging client **104** enables different fashion items and brands to enhance an emotional experience of the user through music or audio generation.

System Architecture

FIG. **2** is a block diagram illustrating further details regarding the messaging system **100**, according to some examples. Specifically, the messaging system **100** is shown to comprise the messaging client **104** and the application servers **114**. The messaging system **100** embodies a number of subsystems, which are supported on the client side by the messaging client **104** and on the sever side by the application servers **114**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **208**, a map system **210**, a game system **212**, and an external resource system **220**.

The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the messaging client **104** and the messaging server **118**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an "event gallery" or an "event story." Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a "story" for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client **104**.

The collection management system **204** further includes a curation interface **206** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **206** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.

The augmentation system **208** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system **208** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The augmentation system **208** operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client **104** based on a geolocation of the client device **102**. In another example, the augmentation system **208** operatively supplies a media overlay to the messaging client **104** based on other information, such as social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay may include text, a graphical element, or image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **208** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **126** and accessed through the database server **120**.

In some examples, the augmentation system **208** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **208** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In other examples, the augmentation system **208** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **208** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time. The augmentation system **208** communicates with the image processing server **122** to obtain AR experiences and presents identifiers of such experiences in one or more user interfaces (e.g., as icons over a real-time image or video or as thumbnails or icons in interfaces dedicated for presented identifiers of AR experiences). Once an AR experience is selected, one or more images, videos, or AR graphical elements are retrieved and presented as an overlay on top of the images or video captured by the client device **102**. In some cases, the camera is switched to a front-facing view (e.g., the front-facing camera of the client device **102** is activated in response to activation of a particular AR experience) and the images from the front-facing camera of the client device **102** start being displayed on the client device **102** instead of the rear-facing camera of the client device **102**. The one or more images, videos, or AR graphical elements are retrieved and presented as an overlay on top of the images that are captured and displayed by the front-facing camera of the client device **102**.

In other examples, the augmentation system **208** is able to communicate and exchange data with another augmentation system **208** on another client device **102** and with the server via the network **112**. The data exchanged can include a session identifier that identifies the shared AR session, a transformation between a first client device **102** and a second client device **102** (e.g., a plurality of client devices **102** include the first and second devices) that is used to align the shared AR session to a common point of origin, a common coordinate frame, functions (e.g., commands to invoke functions), and other payload data (e.g., text, audio, video or other multimedia data).

The augmentation system **208** sends the transformation to the second client device **102** so that the second client device **102** can adjust the AR coordinate system based on the transformation. In this way, the first and second client devices **102** synch up their coordinate systems and frames for displaying content in the AR session. Specifically, the augmentation system **208** computes the point of origin of the second client device **102** in the coordinate system of the first client device **102**. The augmentation system **208** can then determine an offset in the coordinate system of the second client device **102** based on the position of the point of origin from the perspective of the second client device **102** in the coordinate system of the second client device **102**. This offset is used to generate the transformation so that the second client device **102** generates AR content according to a common coordinate system or frame as the first client device **102**.

The augmentation system **208** can communicate with the client device **102** to establish individual or shared AR sessions. The augmentation system **208** can also be coupled to the messaging server **118** to establish an electronic group communication session (e.g., group chat, instant messaging) for the client devices **102** in a shared AR session. The electronic group communication session can be associated with a session identifier provided by the client devices **102** to gain access to the electronic group communication session and to the shared AR session. In one example, the client devices **102** first gain access to the electronic group communication session and then obtain the session identifier in the electronic group communication session that allows the client devices **102** to access the shared AR session. In some examples, the client devices **102** are able to access the shared AR session without aid or communication with the augmentation system **208** in the application servers **114**.

The map system **210** provides various geographic location functions and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **210** enables the display of user icons or avatars (e.g., stored in profile data **316**) on a map to indicate a current or past location of "friends" of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to "friends" of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

The game system **212** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games (e.g., web-based games or web-based applications) that can be launched by a user within the context of the messaging client **104** and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

The external resource system **220** provides an interface for the messaging client **104** to communicate with external app(s) servers **110** to launch or access external resources. Each external resource (apps) server **110** hosts, for example, a markup language (e.g., HTML5) based application or small-scale version of an external application (e.g., game, utility, payment, or ride-sharing application that is external to the messaging client **104**). The messaging client **104** may launch a web-based resource (e.g., application) by accessing the HTML5 file from the external resource (apps) servers **110** associated with the web-based resource. In certain examples, applications hosted by external resource servers **110** are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the messaging server **118**. The SDK includes APIs with functions that can be called or invoked by the web-based application. In certain examples, the messaging server **118** includes a JavaScript library that provides a given third-party resource access to certain user data of the messaging client **104**. HTML5 is used as an example technology for programming games, but applications and resources programmed based on other technologies can be used.

In order to integrate the functions of the SDK into the web-based resource, the SDK is downloaded by an external resource (apps) server **110** from the messaging server **118** or is otherwise received by the external resource (apps) server **110**. Once downloaded or received, the SDK is included as part of the application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the messaging client **104** into the web-based resource.

The SDK stored on the messaging server **118** effectively provides the bridge between an external resource (e.g., third-party or external applications **109** or applets and the messaging client **104**). This provides the user with a seamless experience of communicating with other users on the messaging client **104**, while also preserving the look and feel of the messaging client **104**. To bridge communications between an external resource and a messaging client **104**, in certain examples, the SDK facilitates communication between external resource servers **110** and the messaging client **104**. In certain examples, a Web ViewJavaScript-Bridge running on a client device **102** establishes two one-way communication channels between an external resource and the messaging client **104**. Messages are sent between the external resource and the messaging client **104** via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

By using the SDK, not all information from the messaging client **104** is shared with external resource servers **110**. The SDK limits which information is shared based on the needs of the external resource. In certain examples, each external resource server **110** provides an HTML5 file corresponding to the web-based external resource to the messaging server **118**. The messaging server **118** can add a visual representation (such as a box art or other graphic) of the web-based external resource in the messaging client **104**. Once the user selects the visual representation or instructs the messaging client **104** through a graphical user interface of the messaging client **104** to access features of the web-based external resource, the messaging client **104** obtains the HTML5 file and instantiates the resources necessary to access the features of the web-based external resource.

The messaging client **104** presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the messaging client **104** determines whether the launched external resource has been previously authorized to access user data of the messaging client **104**. In response to determining that the launched external resource has been previously authorized to access user data of the messaging client **104**, the messaging client **104** presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the messaging client **104**, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the messaging client **104** slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle of or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the messaging client **104** adds the external resource to a list of authorized external resources and allows the external resource to access user data from the messaging client **104**. In some examples, the external resource is authorized by the messaging client **104** to access the user data in accordance with an OAuth 2 framework.

The messaging client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale external applications (e.g., a third-party or external application **109**) are provided with access to a first type of user data (e.g., only 2D avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of external applications (e.g., web-based versions of third-party applications) are provided with access to a second type of user data (e.g., payment information, 2D avatars of users, 3D avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

An AR fashion control system **224** segments a fashion item, such as a shirt, worn by a user depicted in an image (or video) or multiple fashion items worn respectively by multiple users depicted in an image (or video). An illustrative implementation of the AR fashion control system **224** is shown and described in connection with FIG. **5** below.

Specifically, the AR fashion control system **224** is a component that can be accessed by an AR/VR application implemented on the client device **102**. The AR/VR application uses an RGB camera to capture a monocular image of a user and the garment or garments (alternatively referred to as fashion item(s)) worn by the user. The AR/VR application applies various trained machine learning techniques on the captured image of the user wearing the garment to segment the garment (e.g., a shirt, jacket, pants, dress, and so forth) worn by the user in the image and to apply one or more AR visual effects (e.g., game-based AR elements) to the captured image. Segmenting the garment results in an outline of the borders of the garment that appear in the image or video. Pixels within the borders of the segmented garment correspond to the garment or clothing worn by the user. The segmented garment is used to distinguish the clothing or garment worn by the user from other objects or elements depicted in the image, such as parts of the user's body (e.g., arms, head, legs, and so forth) and the background of the image which can be separately segmented and tracked. In some implementations, the AR/VR application continuously captures images of the user wearing the garment in real time or periodically to continuously or periodically update the applied one or more visual effects. This allows the user to move around in the real world and see the one or more visual effects update in real time.

In order for the AR/VR application to apply the one or more visual effects directly from a captured RGB image, the AR/VR application obtains a trained machine learning technique from the AR fashion control system **224**. The trained machine learning technique processes the captured RGB image to generate a segmentation from the captured image that corresponds to the garment worn by the user(s) depicted in the captured RGB image.

In training, the AR fashion control system **224** obtains a first plurality of input training images that include depictions of one or more users wearing different garments. These training images also provide the ground truth information about the segmentations of the garments worn by the users depicted in each image. A machine learning technique (e.g., a deep neural network) is trained based on features of the plurality of training images. Specifically, the first machine learning technique extracts one or more features from a given training image and estimates a segmentation of the garment worn by the user depicted in the given training image. The machine learning technique obtains the ground truth information corresponding to the training image and adjusts or updates one or more coefficients or parameters to improve subsequent estimations of segmentations of the garment, such as the shirt.

Data Architecture

FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the messaging server system **108**, according to certain examples. While the content of the database **126** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **126** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **302**, are described below with reference to FIG. **4**.

An entity table **306** stores entity data, and is linked (e.g., referentially) to an entity graph **308** and profile data **316**. Entities for which records are maintained within the entity table **306** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph **308** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization), interested-based, or activity-based, merely for example.

The profile data **316** stores multiple types of profile data about a particular entity. The profile data **316** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **316** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), and a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100** and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

Where the entity is a group, the profile data **316** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

The database **126** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **304**) and images (for which data is stored in an image table **312**).

The database **126** can also store data pertaining to individual and shared AR sessions. This data can include data communicated between an AR session client controller of a first client device **102** and another AR session client controller of a second client device **102**, and data communicated between the AR session client controller and the augmentation system **208**. Data can include data used to establish the common coordinate frame of the shared AR scene, the transformation between the devices, the session identifier, images depicting a body, skeletal joint positions, wrist joint positions, feet, and so forth.

Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102**, or the current time.

Other augmentation data that may be stored within the image table **312** includes AR content items (e.g., corresponding to applying AR experiences). An AR content item or AR item may be a real-time special effect and sound that may be added to an image or a video.

As described above, augmentation data includes AR content items, overlays, image transformations, AR images, AR logos or emblems, and similar terms that refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device **102** and then displayed on a screen of the client device **102** with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device **102** with access to multiple AR content items, a user can use a single video clip with multiple AR content items to see how the different AR content items will modify the stored clip. For example, multiple AR content items that apply different pseudorandom movement models can be applied to the same content by selecting different AR content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device **102** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different AR content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

Data and various systems using AR content items or other such transform systems to modify content using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various examples, different methods for achieving such transformations may be used. Some examples may involve generating a 3D mesh model of the object or objects and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be 2D or 3D) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). AR content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of an object's elements, characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh is used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A first set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request for modification, thus transforming the frames of the video stream. Depending on the specific request for modification, properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas that are based on a request for modification; and modifying or distorting the elements of an area or object. In various examples, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an ASM algorithm is applied to the face region of an image to detect facial feature reference points.

Other methods and algorithms suitable for face detection can be used. For example, in some examples, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

In some examples, a search is started for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a messaging client **104** operating on the client device **102**. The transformation system operating within the messaging client **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transformation system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the client device **102** as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured and the selected modification icon remains toggled. Machine-taught neural networks may be used to enable such modifications.

The graphical user interface, presenting the modification performed by the transformation system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various examples, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some examples, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

A story table **314** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **306**). A user may create a "personal story" in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

A collection may also constitute a "live story," which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a "live story" may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client **104**, based on his or her location. The end result is a "live story" told from a community perspective.

A further type of content collection is known as a "location story," which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific organization or other entity (e.g., is a student on the university campus).

As mentioned above, the video table **304** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. Similarly, the image table **312** stores image data associated with messages for which message data is stored in the entity table **306**. The entity table **306** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **312** and the video table **304**.

Trained machine learning technique(s) **307** stores parameters that have been trained during training of the AR fashion control system **224**. For example, trained machine learning techniques **307** stores the trained parameters of one or more neural network machine learning techniques.

Segmentation training images **309** stores a plurality of images that each depict one or more users wearing different garments. The plurality of images stored in the segmentation training images **309** includes various depictions of one or more users wearing different garments together with segmentations of the garments that indicate which pixels in the images correspond to the garments and which pixels correspond to a background or a user's body parts in the images. Namely the segmentations provide the borders of the garments depicted in the images. These segmentation training images **309** are used by the AR fashion control system **224** to train the machine learning technique used to generate a segmentation of one or more garments depicted in a received RGB monocular image. In some cases, the segmentation training images **309** include ground truth skeletal key points of one or more bodies depicted in the respective training monocular images to enhance segmentation performance on various distinguishing attributes (e.g., shoulder straps, collar, or sleeves) of the garments. In some cases, the segmentation training images **309** include a plurality of image resolutions of bodies depicted in the images. The segmentation training images **309** can include labeled and unlabeled image and video data. The segmentation training images **309** can include a depiction of a whole body of a particular user, an image that lacks a depiction of any user (e.g., a negative image), a depiction of a plurality of users wearing different garments, and depictions of users wearing garments at different distances from an image capture device.

Data Communications Architecture

FIG. **4** is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by a messaging client **104** for communication to a further messaging client **104** or the messaging server **118**. The content of a particular message **400** is used to populate the message table **302** stored within the database **126**, accessible by the messaging server **118**. Similarly, the content of a message **400** is stored in memory as "in-transit" or "in-flight" data of the client device **102** or the application servers **114**. A message **400** is shown to include the following example components:

- message identifier **402**: a unique identifier that identifies the message **400**.
- message text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the message **400**.
- message image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **312**.
- message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the video table **304**.
- message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the message **400**.
- message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data for a sent or received message **400** may be stored in the augmentation table **310**.
- message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the messaging client **104**.
- message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **406**, or a specific video in the message video payload **408**).
- message story identifier **418**: identifier values identifying one or more content collections (e.g., "stories" identified in the story table **314**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple

images within the message image payload **406** may each be associated with multiple content collections using identifier values.

message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** on which the message **400** was generated and from which the message **400** was sent.

message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the message **400** is addressed.

The contents (e.g., values) of the various components of message **400** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **312**. Similarly, values within the message video payload **408** may point to data stored within a video table **304**, values stored within the message augmentation data **412** may point to data stored in an augmentation table **310**, values stored within the message story identifier **418** may point to data stored in a story table **314**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **306**.

AR Fashion Control System

FIG. **5** is a block diagram showing an example AR fashion control system **224**, according to example examples. AR fashion control system **224** includes a set of components **510** that operate on a set of input data (e.g., a monocular image **501** depicting a real body of a user wearing a shirt and upper garment segmentation training image data **502**). The set of input data is obtained from segmentation training images **309** stored in database(s) (FIG. **3**) during the training phases and is obtained from an RGB camera of a client device **102** when an AR/VR application is being used, such as by a messaging client **104**. AR fashion control system **224** includes a machine learning technique module **512**, a skeletal key-points module **511**, an upper garment segmentation module **514**, an image modification module **518**, an AR effect selection module **519**, a 3D body tracking module **513**, a whole-body segmentation module **515**, and an image display module **520**.

During training, the AR fashion control system **224** receives a given training image or video (e.g., monocular image **501** depicting a real body of a user wearing a garment, such as an image of a user wearing a shirt (short sleeve, t-shirt, or long sleeve), jacket, tank top, sweater, and so forth; a lower body garment, such as pants or a skirt; a whole body garment, such as a dress or overcoat; or any suitable combination thereof or depicting multiple users simultaneously wearing respective combinations of upper body garments, lower body garments or whole body garments) from segmentation training image data **502**. The AR fashion control system **224** applies one or more machine learning techniques using the machine learning technique module **512** on the given training image or video. The machine learning technique module **512** extracts one or more features from the given training image or video to estimate a segmentation of the garment(s) worn by the user(s) depicted in the image. For example, the machine learning technique module **512** obtains the given training image or video depicting a user wearing a shirt. The machine learning technique module **512** extracts features from the image and segments or specifies which pixels in the image correspond to the shirt worn by the user and which pixels correspond to a background or correspond to parts of the user's body. Namely, the segmentation output by the machine learning technique module **512** identifies borders of a garment (e.g., the shirt) worn by the user in the given training image.

The machine learning technique module **512** retrieves garment segmentation information associated with the given training image or video. The machine learning technique module **512** compares the estimated segmentation (that can include an identification of multiple garments worn by respective users in the image in case there exist multiple users in the image) with the ground truth garment segmentation provided as part of the segmentation training image data **502**. Based on a difference threshold or deviation of the comparison, the machine learning technique module **512** updates one or more coefficients or parameters and obtains one or more additional segmentation training images or videos. After a specified number of epochs or batches of training images have been processed and/or when the difference threshold or deviation reaches a specified value, the machine learning technique module **512** completes training and the parameters and coefficients of the machine learning technique module **512** are stored in the trained machine learning technique(s) **307**.

In some examples, the machine learning technique module **512** implements multiple segmentation models of the machine learning technique. Each segmentation model of the machine learning technique module **512** may be trained on a different set of training images associated with a specific resolution. Namely, one of the segmentation models can be trained to estimate a garment segmentation for images having a first resolution (or a first range of resolutions). A second of the segmentation models can be trained to estimate a garment segmentation for images having a second resolution (or a second range of resolutions different from the first range of resolutions). In this way, different complexities of the machine learning technique module **512** can be trained and stored. When a given device having certain capabilities uses the AR/VR application, a corresponding one of the various garment segmentation models can be provided to perform the garment segmentation that matches the capabilities of the given device. In some cases, multiple garment segmentation models of each of the machine leaning techniques implemented by the AR fashion control system **224** can be provided, with each configured to operate with a different level of complexity. The appropriate segmentation model(s) with the appropriate level of complexity can then be provided to a client device **102** for segmenting garments depicted in one or more images.

In some examples, during training, the machine learning technique module **512** receives 2D skeletal joint information from a skeletal key-points module **511**. The skeletal key-points module **511** tracks skeletal key points of a user depicted in a given training image (e.g., head joint, shoulder joints, hip joints, leg joints, and so forth) and provides the 2D or 3D coordinates of the skeletal key points. This information is used by the machine learning technique module **512** to identify distinguishing attributes of the garment depicted in the training image.

The garment segmentation generated by the machine learning technique module **512** is provided to the upper garment segmentation module **514**. The upper garment segmentation module **514** can determine that the elbow joint output by the skeletal key-points module **511** is at a position that is within a threshold distance away from a given edge of the border of the shirt garment segmentation. In response, the upper garment segmentation module **514** can determine that the garment corresponds to a t-shirt or short sleeve shirt and that the given edge corresponds to a sleeve of the shirt. In such circumstances, the upper garment segmentation module **514** can adjust weights of the parameters or the loss function used to update parameters of the machine learning technique module **512** to improve segmentation of upper body garments, such as shirts. More specifically, the upper garment segmentation module **514** can determine that a given distinguishing attribute is present in the garment segmentation that is generated based on a comparison of skeletal joint positions to borders of the garment segmentation. In such circumstances, the upper garment segmentation module **514** adjusts the loss function or weights used to update the parameters of the machine learning technique module **512** for the training image depicting the garment with the distinguishing attribute. Similarly, the upper garment segmentation module **514** can adjust the loss or the parameter weights based on a difference between the garment segmentation and the pixels corresponding to the background of the image.

The upper garment segmentation module **514** is used to track a 2D or 3D position of the segmented shirt in subsequent frames of a video. This enables one or more AR elements to be displayed on the shirt and be maintained at their respective positions on the shirt as the position of the shirt moves around the screen. In this way, the upper garment segmentation module **514** can determine and track which portions of the shirt are currently shown in the image that depicts the user and to selectively adjust the corresponding AR elements that are displayed. For example, a given AR element can be displayed on a left sleeve of the shirt in a first frame of the video. The upper garment segmentation module **514** can determine that the user has turned left in a second frame of the video, meaning that the left sleeve no longer appears in the second frame. In response, the upper garment segmentation module **514** can omit the whole or a portion of the given AR element that was displayed on the left sleeve of the shirt.

After training, AR fashion control system **224** receives an input image **501** (e.g., monocular image depicting a user wearing a garment or multiple users wearing respective garments) as a single RGB image from a client device **102**. The AR fashion control system **224** applies the trained machine learning technique module **512** to the received input image **501** to extract one or more features of the image to generate a segmentation of the garment or garments depicted in the image **501**. This segmentation is provided to the upper garment segmentation module **514** to track the 2D or 3D position of the shirt (upper garment) in the current frame of the video and in subsequent frames.

FIG. **6** is a diagrammatic representation of outputs of the AR fashion control system **224**, in accordance with some examples. Specifically, FIG. **6** shows a garment segmentation **600** generated by the upper garment segmentation module **514**. In one example, the upper garment segmentation module **514** generates a first garment segmentation **612** representing pixel locations of a shirt (upper garment) worn by a user. In another example, the upper garment segmentation module **514** generates a second garment segmentation representing pixel locations of a short sleeve shirt (upper garment) worn by a user. In another example, the upper garment segmentation module **514** generates a third garment segmentation representing pixel locations of a jacket (upper garment) (not shown) worn by a user.

Referring back to FIG. **5**, AR effect selection module **519** receives a selection of an AR gaming application in association with a person depicted in an image or video captured by a client device **102**. Based on the selected AR gaming application, the AR effect selection module **519** selects and applies one or more game-based AR elements to the fashion item (e.g., shirt) segmentation received from the upper garment segmentation module **514**. This fashion item segmentation combined with the one or more game-based AR elements is provided to the image modification module **518** to render an image or video that depicts the user wearing a shirt or fashion item with the one or more game-based AR elements.

For example, a user of the AR/VR application may be presented with an option to select an AR gaming application or experience to control display of game-based AR elements on a shirt worn by the user. In response to receiving a user selection of the option, a camera (e.g., front-facing or rear-facing camera) is activated to begin capturing an image or video of the user wearing a shirt (or upper garment or fashion item). The image or video depicting the user wearing the shirt (or upper garment or fashion item) is provided to the AR effect selection module **519** to apply one or more game-based AR elements to the shirt. The AR effect selection module **519** selects between various applications/modifications of game-based AR elements displayed on the shirt (or upper garment or fashion item) worn by the user, such as based on gestures or movement of the user detected by the 3D body tracking module **513** and/or whole-body segmentation module **515**. FIGS. **7** and **8** show illustrative outputs of one or more of the visual effects that can be selected and applied by the AR effect selection module **519**.

The image modification module **518** can adjust the image captured by the camera based on the game-based AR effect selected by the visual effect selection module **519**. The image modification module **518** adjusts the way in which the garment(s) worn by the user is/are presented in an image or video, such as by changing the color or occlusion pattern of the garment worn by the user based on the garment segmentation and applying one or more game-based AR elements to the fashion item worn by the user depicted in the image or video. Image display module **520** combines the adjustments made by the image modification module **518** into the received monocular image or video depicting the user's body. The image or video is provided by the image display module **520** to the client device **102** and can then be sent to another user or stored for later access and display.

In some examples, the image modification module **518** receives 3D body tracking information representing the 3D positions of the user depicted in the image from the 3D body tracking module **513**. The 3D body tracking module **513** generates the 3D body tracking information by processing the image **501** using additional machine learning techniques. The image modification module **518** can also receive a whole-body segmentation representing which pixels in the image correspond to the whole body of the user from another machine learning technique. The whole-body segmentation can be received from the whole-body segmentation module **515**. The whole-body segmentation module **515** generates the whole-body segmentation by processing the image **501** using a machine learning technique.

The image modification module **518** can control the display of virtual or game-based AR elements based on the garment segmentation provided by the upper garment segmentation module **514** and based on the 3D body tracking positions of the user and the whole-body segmentation of the user. The image modification module **518** and/or the AR effect selection module **519** can control the progress of the gaming application based on how the game-based AR elements are adjusted in response to movement of the user's body. The progress of the gaming application can represent an objective or the game and can determine a score of the user in the game. Specifically, the image modification module **518** can control the occlusion pattern of an AR element relative to the real-world garment corresponding to the garment segmentation. Namely, the image modification module **518** determines which portion of the AR element to occlude with pixels of the real-world garment and/or also determines which portion of the real-world garment pixels to occlude with pixels of the AR element. The image modification module **518** can control how different game-based AR elements that are displayed on the fashion item worn by the user depicted in the image or video are moved around or are transitioned based on movement of the user. The image modification module **518** can also determine contact between a user's body parts depicted in the image or video relative to the game-based AR elements that are displayed on the fashion item worn by the user depicted in the image or video or that are moved towards the fashion item.

In an example, the upper garment segmentation module **514** receives a 3D model of the person depicted in an image or video from the 3D body tracking module **513**. The upper garment segmentation module **514** identifies or outlines a portion of the 3D model based on the segmentation of the upper garment. For example, the upper garment segmentation module **514** can generate a cutout of a portion of the 3D model that corresponds to the upper garment segmentation. In an implementation, the upper garment segmentation module **514** can identify a collection of pixels within the 3D model of the person depicted in the image or video that correspond to the position of the upper garment segmentation. Namely, the upper garment segmentation module **514** can overlay the upper garment segmentation on top of the 3D model of the person and can map that overlapped portion to the fashion item being worn by the person or user depicted in the image or video. In this way, the upper garment segmentation module **514** can track exclusively movement of the portion of the 3D model of the person depicted in the image or video that corresponds to the fashion item worn by the person or user.

The upper garment segmentation module **514** can determine changes to a position of the portion of the 3D model that has been outlined (identified as corresponding to the upper garment segmentation). Based on the changes, the upper garment segmentation module **514** can update a display position of AR elements that are displayed on the fashion item worn by the person or user depicted in the image or video. For example, the upper garment segmentation module **514** can communicate with an AR effect selection module **519** to display and update one or more AR elements on a fashion item worn by the person or user depicted in the image or video. The AR effect selection module **519** can display the one or more AR elements on a first portion of the fashion item in a first frame of the video while the person is positioned at a first location in the first frame. In one implementation, the AR effect selection module **519** can display the one or more AR elements on a shirt pocket of the person or user depicted in the image or video. The AR effect selection module **519** can determine that the person has moved from the first location to a second location in a second frame of the video, which causes the first portion of the fashion item to be moved to a new location in the second frame of the video. The AR effect selection module **519** can determine the movement from the first location to the second location based on changes to a position of the 3D model that has been outlined as corresponding to the upper garment segmentation. In response, the AR effect selection module **519** can update a display position of the one or more AR elements in the second frame to maintain the display of the one or more AR elements on the first portion of the fashion item. In this way, the one or more AR elements continue to be displayed on top of the shirt pocket as the person moves from the first location to the second location in the video.

In another example, the AR effect selection module **519** can determine that the portion of the 3D model corresponding to the upper garment segmentation has been rotated by a specified amount in the video depicting the person or user. In such cases, the AR effect selection module **519** rotates the one or more AR elements by the specified amount in response to determining that the portion of the 3D model has been rotated by a specified amount in the video. As another example, the AR elements can be displayed on a given portion of the fashion item worn by the person or user in the image or video. The AR effect selection module **519** can identify the given portion of the fashion item on the portion of the 3D model and can determine that the portion of the 3D model has been rotated by a specified amount in a second frame. In response, the AR effect selection module **519** can determine that the portion of the fashion item fails to appear in the second frame. For example, the AR element may have been displayed on a left shoulder of the person. When the person turns left relative to the camera, the left shoulder no longer appears. As a result, the AR effect selection module **519** can remove at least a portion of the one or more AR elements from being displayed.

In another example, the upper garment segmentation module **514** can generate a 2D representation of the fashion item in a UV coordinate space. The upper garment segmentation module **514** can determine a mapping of pixels of the fashion item in the UV coordinate space to pixels of the portion of the 3D model in the UV coordinate space. For example, the upper garment segmentation module **514** can overlay the upper garment segmentation in the UV coordinate space on a UV representation of the 3D model to identify the pixels of the 3D model in the UV coordinate space that correspond to the upper garment segmentation. The upper garment segmentation module **514** can determine changes to a position of the pixels in the UV coordinate space of the 3D model. Based on the changes, the upper garment segmentation module **514** can update a display position of AR elements that are displayed on the fashion item worn by the person or user depicted in the image or video.

In one example, as shown in FIG. **7**, the AR effect selection module **519** can apply one or more AR element **730** to a shirt **710** worn by a user **720** depicted in an image **700** captured by a client device **102**. The one or more AR element **730** can include an AR game board, an AR gaming controller for controlling a gaming application interface, an AR ball game, an AR capture game, an AR flying saucer game, or any other type of AR gaming experience that can be displayed on a fashion item of a user and interacted by gestures performed by the user. Other types of AR element **730** can include AR music related AR elements (discussed above), AR avatars, AR voice transcriptions, AR elements that are based on voice expressions of the user **720**, and the like.

In an example, the AR effect selection module **519** presents an AR drawing board as the one or more AR element **730** on the fashion item, such as the shirt **710** worn by the person or user **720** depicted in the image **700**. The AR effect selection module **519** can detect movement of a body part of the person depicted in the image **700** or video. In response to detecting that the body part overlaps the AR drawing board, the AR effect selection module **519** adds one or more AR graphics to the AR drawing board. For example, the AR effect selection module **519** can draw or paint a line on the AR drawing board along a direction of the body part that is detected to move over the position of the AR drawing board. The AR effect selection module **519** can update display positions of the AR element **730** displayed on the shirt as the user **720** moves around the video.

For example, the AR element **730** can be displayed on a first portion of the shirt **710** when the user **720** is in a first position in a first frame of a video. The AR effect selection module **519** can determine that a portion of the 3D model of the user **720** corresponding to the shirt **710** (which has been cut out or outlined based on the segmentation of the shirt **710**) has been moved to a second position in a second frame of the video. In response, the AR effect selection module **519** can compute the trajectory or path of movement and can move the AR element **730** along the same trajectory or path. The AR effect selection module **519** can also adjust a scale of the AR element **730** based on changes to a scale of the portion of the 3D model of the user **720** corresponding to the shirt **710**.

As another example shown in FIG. **8**, the AR effect selection module **519** can generate an avatar **840** for display on the shirt **810** (or fashion item) worn by the person or user **826** depicted in the image **800** or video. The AR effect selection module **519** can select a random pose for the avatar **840** or can access a list of avatar poses associated with an AR experience. The avatar **840** can be moved around in the video based on movement of the user **826** by tracking changes to display positions of a portion of a 3D model corresponding to a segmentation of the shirt **810**. For example, the avatar **840** can be maintained in a particular position and orientation relative to the shirt **810** as the user **826** changes poses and positions or moves around. For example, if the user **826** leans towards the left by a certain degree, the portion of the 3D model of the user **826** similarly is determined to lean by the certain degree to the left. In response, the AR effect selection module **519** can rotate the avatar **840** by the certain degree to the left to maintain the orientation and position of the avatar **840** relative to movement of the shirt **810**. This makes it appear as though the avatar **840** is part of the real-world shirt **810**. The AR effect selection module **519** can determine that a first portion of the avatar **840** is displayed on a first portion of the 3D model and a second portion of the avatar **840** is displayed on a second portion of the 3D model. The AR effect selection module **519** can determine that, as a result of the user **826** twisting or turning in a given direction, the first portion of the 3D model is no longer visible but the second portion remains visible. As a result, the AR effect selection module **519** can maintain a display of the second portion of the avatar **840** and remove from display the first portion of the avatar **840**.

As another example, the AR effect selection module **519** can display an AR object on the shirt **810** worn by the user **826**. The AR object can have a top portion. The top portion can, in a first frame of a video, be pointing towards a top of the display while the user **826** is standing up. The AR effect selection module **519** can determine that a portion of the 3D model in a second frame of the video has changed orientation as a result of the user **826** laying down on his/her side. Namely, tops of shoulders of the user **826** can be pointing towards a top of the display in the first frame and after the user lays down, the tops of the shoulders of the user **826** can be pointing towards a right edge of the display. This causes the portion of the 3D model (that has been outlined based on a segmentation of the shirt **810**) to be changed from being oriented in one direction to being rotated 90 degrees and oriented along a second direction. In response, the AR effect selection module **519** can update the AR object to similarly be rotated by 90 degrees such that the top portion of the AR object is changed from being pointing towards the top of the display in the first frame to pointing towards the right edge of the display in the second frame. Other changes in 3D axes determined to be made to the 3D model portion corresponding to the segmentation of the shirt **810** can be similarly applied to the AR object.

In one implementation, the AR effect selection module **519** can allow the user **826** to purchase a given AR element and can associate a non-fungible token with the given AR element in response to the user **826** completing a purchase transaction for the given AR element. For example, the AR effect selection module **519** can detect a user input that selects a buy shirt option **850**. In response, the AR effect selection module **519** can capture a screenshot or a video that includes the fashion item (e.g., shirt **810**) worn by the user **826** depicted in the image or video and that includes or is combined with the one or more AR elements (e.g., the avatar **840**, or any other AR element discussed above). The AR effect selection module **519** can generate an AR element that represents or includes the fashion item worn by the user **826** depicted in the image **800** or video combined with the one or more AR elements. The AR effect selection module **519** can generate a non-fungible token and associate the non-fungible token with the AR element. This allows the user **826** or person to exchange or sell the AR element as a unique asset with one or more other users.

FIG. **9** is a flowchart of a process **900** performed by the AR fashion control system **224**, in accordance with some example examples. Although the flowchart can describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part, may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.

At operation **901**, the AR fashion control system **224** (e.g., a client device **102** or a server) receives a video that includes a depiction of a person wearing a fashion item, as discussed above.

At operation **902**, the AR fashion control system **224** generates a 3D model of the person depicted in the video, as discussed above.

At operation **903**, the AR fashion control system **224** generates a segmentation of the fashion item worn by the person depicted in the video, as discussed above.

At operation **904**, the AR fashion control system **224** outlines a portion of the 3D model based on the segmentation of the fashion item, as discussed above.

At operation **905**, the AR fashion control system **224** tracks movement of the portion of the 3D model in the video, as discussed above.

At operation **906**, the AR fashion control system **224**, in response to tracking the movement of the portion of the 3D model in the video, modifies a display position of one or more AR elements on the fashion item in the video, as discussed above.

Machine Architecture

FIG. **10** is a diagrammatic representation of the machine **1000** within which instructions **1008** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1008** may cause the machine **1000** to execute any one or more of the methods described herein. The instructions **1008** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. The machine **1000** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1008**, sequentially or otherwise, that specify actions to be taken by the machine **1000**. Further, while only a single machine **1000** is illustrated, the term "machine" shall also be taken to include a collection of machines that individually or jointly execute the instructions **1008** to perform any one or more of the methodologies discussed herein. The machine **1000**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1000** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

The machine **1000** may include processors **1002**, memory **1004**, and input/output (I/O) components **1038**, which may be configured to communicate with each other via a bus **1040**. In an example, the processors **1002** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1006** and a processor **1010** that execute the instructions **1008**. The term "processor" is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as "cores") that may execute instructions contemporaneously. Although FIG. **10** shows multiple processors **1002**, the machine **1000** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory **1004** includes a main memory **1012**, a static memory **1014**, and a storage unit **1016**, all accessible to the processors **1002** via the bus **1040**. The main memory **1004**, the static memory **1014**, and the storage unit **1016** store the instructions **1008** embodying any one or more of the methodologies or functions described herein. The instructions **1008** may also reside, completely or partially, within the main memory **1012**, within the static memory **1014**, within machine-readable medium **1018** within the storage unit **1016**, within at least one of the processors **1002** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**.

The I/O components **1038** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1038** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1038** may include many other components that are not shown in FIG. **10**. In various examples, the I/O components **1038** may include user output components **1024** and user input components **1026**. The user output components **1024** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1026** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components **1038** may include biometric components **1028**, motion components **1030**, environmental components **1032**, or position components **1034**, among a wide array of other components. For example, the biometric components **1028** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1030** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components **1032** include, for example, one or more cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., "selfies"), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad, or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

The position components **1034** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1038** further include communication components **1036** operable to couple the machine **1000** to a network **1020** or devices **1022** via respective coupling or connections. For example, the communication components **1036** may include a network interface component or another suitable device to interface with the network **1020**. In further examples, the communication components **1036** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1022** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components **1036** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1036** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1036**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory **1012**, static memory **1014**, and memory of the processors **1002**) and storage unit **1016** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1008**), when executed by processors **1002**, cause various operations to implement the disclosed examples.

The instructions **1008** may be transmitted or received over the network **1020**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1036**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1008** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1022**.

Software Architecture

FIG. **11** is a block diagram **1100** illustrating a software architecture **1104**, which can be installed on any one or more of the devices described herein. The software architecture **1104** is supported by hardware such as a machine **1102** that includes processors **1120**, memory **1126**, and I/O components **1138**. In this example, the software architecture **1104** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1104** includes layers such as an operating system **1112**, libraries **1110**, frameworks **1108**, and applications **1106**. Operationally, the applications **1106** invoke API calls **1150** through the software stack and receive messages **1152** in response to the API calls **1150**.

The operating system **1112** manages hardware resources and provides common services. The operating system **1112** includes, for example, a kernel **1114**, services **1116**, and drivers **1122**. The kernel **1114** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1114** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **1116** can provide other common services for the other software layers. The drivers **1122** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1122** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries **1110** provide a common low-level infrastructure used by applications **1106**. The libraries **1110** can include system libraries **1118** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1110** can include API libraries **1124** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1110** can also include a wide variety of other libraries **1128** to provide many other APIs to the applications **1106**.

The frameworks **1108** provide a common high-level infrastructure that is used by the applications **1106**. For example, the frameworks **1108** provide various graphical user interface functions, high-level resource management, and high-level location services. The frameworks **1108** can provide a broad spectrum of other APIs that can be used by the applications **1106**, some of which may be specific to a particular operating system or platform.

In an example, the applications **1106** may include a home application **1136**, a contacts application **1130**, a browser application **1132**, a book reader application **1134**, a location application **1142**, a media application **1144**, a messaging application **1146**, a game application **1148**, and a broad assortment of other applications such as an external application **1140**. The applications **1106** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1106**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the external application **1140** (e.g., an application developed using the ANDROID™ or IOS™ SDK by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the external application **1140** can invoke the API calls **1150** provided by the operating system **1112** to facilitate functionality described herein.

Glossary

"Carrier signal" refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

"Client device" refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistant (PDA), smartphones, tablet, ultrabook, netbook, laptop, multi-processor system, microprocessor-based or programmable consumer electronics, game console, set-top box, or any other communication device that a user may use to access a network.

"Communication network" refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

"Component" refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions.

Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A "hardware component" is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an ASIC. A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase "hardware component" (or "hardware-implemented component") should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time.

Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices and can operate on a resource (e.g., a collection of information).

The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, "processor-implemented component" refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors **1002** or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a "cloud computing" environment or as a "software as a service" (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

"Computer-readable storage medium" refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms "machine-readable medium," "computer-readable medium," and "device-readable medium" mean the same thing and may be used interchangeably in this disclosure.

"Ephemeral message" refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video, and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

"Machine storage medium" refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines, and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media, and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms "machine-storage medium," "device-storage medium," and "computer-storage medium" mean the same thing and may be used interchangeably in this disclosure. The terms "machine-storage media," "computer-storage media," and "device-storage media" specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term "signal medium."

"Non-transitory computer-readable storage medium" refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

"Signal medium" refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term "signal medium" shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term "modulated data signal" means a signal that has one or more of its characteristics set or changed in such a matter as to encode information in the signal. The terms "transmission medium" and "signal medium" mean the same thing and may be used interchangeably in this disclosure.

Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

What is claimed is:

1. A method comprising:

receiving, by one or more processors of a device, a video that includes a depiction of an object;

generating, by the one or more processors, a three-dimensional (3D) model of the object;

causing a plurality of augmented reality (AR) objects to appear as coming out of a first portion of the object relative to a surface normal of the first portion of the object along different directions and at different rates; and

using the 3D model of the object to adjust a score based on detected contact between an individual AR object of the plurality of AR objects and a second portion of the object.

2. The method of claim 1, the object comprising a person, the first portion of the object comprising a fashion item, further comprising:

applying a trained machine learning model to the video that includes the depiction of the person wearing the fashion item, the trained machine learning model extracting one or more features from the video, and the trained machine learning model generating a segmentation of the fashion item worn by the person depicted in the video using the extracted one or more features;

outlining a portion of the 3D model based on the segmentation of the fashion item;

tracking movement of the portion of the 3D model in the video to determine that a portion of the fashion item has been moved towards a particular side; and

in response to tracking the movement of the portion of the 3D model in the video, modifying a display position of one or more augmented reality (AR) elements on the fashion item in the video.

3. The method of claim 1, further comprising:

modifying a shape of the one or more AR objects in response to detecting a pose of the object.

4. The method of claim 1, wherein the object comprises a person, the first portion comprises a fashion item worn by the person, and the second portion comprises a body part of the person.

5. The method of claim 1, wherein the one or more AR objects are associated with a non-fungible token, wherein the non-fungible token is associated with the one or more AR objects in response to a purchase transaction.

6. The method of claim 1, further comprising:

generating an AR object comprising the first portion combined with the one or more AR objects; and

generating a non-fungible token associated with the AR object.

7. The method of claim 1, further comprising maintaining a location of the one or more AR objects on the first portion in the video as the object moves around the video.

8. The method of claim 7, further comprising:

displaying the one or more AR objects on the first portion in a first frame of the video, wherein the object is positioned at a first location in the first frame;

determining that the object has moved from the first location to a second location in a second frame of the video, the first portion being moved to a new location in the second frame of the video; and

updating a display position of the one or more AR objects in the second frame to maintain the display of the one or more AR objects on the first portion.

9. The method of claim 8, wherein the first portion comprises a shirt pocket, wherein the one or more AR objects continue to be displayed on top of the shirt pocket as the object moves from the first location to the second location in the video.

10. The method of claim 1, further comprising:

selecting a random pose for an avatar;

causing the avatar with the random pose to be presented to the object;

presenting a timer counting towards a specified value; and

instructing the object to perform a gesture that mirrors the random pose of the avatar before the timer reaches the specified value.

11. The method of claim 10, wherein the one or more AR objects are displayed on a portion of a fashion item, further comprising:

identifying the portion of the fashion item on the portion of the 3D model;

determining that the portion of the 3D model has been rotated by a specified amount in a second frame of the video;

in response to determining that the 3D model has been rotated by the specified amount, determining that the portion of the fashion item fails to appear in the second frame; and

removing at least a portion of the one or more AR objects in response to determining that the portion of the fashion item fails to appear in the second frame.

12. A system comprising:

at least one processor of a device; and

a memory component having instructions stored thereon that, when executed by the at least one processor, cause the at least one processor to perform operations comprising:

receiving a video that includes a depiction of an object;

generating a three-dimensional (3D) model of the object;

causing a plurality of augmented reality (AR) objects to appear as coming out of a first portion of the object relative to a surface normal of the first portion of the object along different directions and at different rates; and

using the 3D model of the object to adjust a score based on detected contact between an individual AR object of the plurality of AR objects and a second portion of the object.

13. The system of claim 12, the operations further comprising:

generating, as the plurality of AR objects, one or more game-based AR game boards;

generating an avatar associated with the one or more game-based AR game boards; and

displaying the one or more game-based AR game boards and the avatar.

14. The system of claim 12, the operations further comprising:

modifying a shape of the plurality of AR objects in response to detecting a pose of the object depicted in the video.

15. The system of claim 12, the operations further comprising:

detecting overlap between a body part of a person comprising the object and a first portion of a fashion item.

16. The system of claim 12, wherein the object comprises a person, the first portion comprises a fashion item worn by the person, and the second portion comprises a body part of the person.

17. The system of claim 16, the operations further comprising:

adjusting a display of the plurality of AR objects based on wrinkles detected on the fashion item.

18. The system of claim 16, the operations further comprising:

receiving, through a microphone, a voice input associated with the person wearing the fashion item in the video;

in response to receiving the voice input, applying a second trained machine learning model to a characteristic of the voice input to detect an emotion associated with the voice input;

generating, as the plurality of AR objects on the fashion item in the video, an avatar having an avatar facial expression;

searching a database of avatar facial expressions that are associated with the emotion associated with the voice input, and adjusting the avatar facial expression to correspond to the characteristic of the voice input associated with the person depicted in the video.

19. A non-transitory computer-readable storage medium having stored thereon instructions that, when executed by at least one processor of a device, cause the at least one processor to perform operations comprising:

receiving a video that includes a depiction of an object;

generating a three-dimensional (3D) model of the object;

causing a plurality of augmented reality (AR) objects to appear as coming out of a first portion of the object relative to a surface normal of the first portion of the object along different directions and at different rates; and

using the 3D model of the object to adjust a score based on detected contact between an individual AR object of the plurality of AR objects and a second portion of the object.

20. The non-transitory computer-readable storage medium of claim **19**, the operations comprising:

modifying a shape of the plurality of AR objects in response to detecting a pose of the object.

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